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HEART

Anatomy and physiology

The heart comprises two muscular pumps working in series, covered in a serous sac (pericardium) that allows free movement with each heart beat and respiration (Fig. 4.1). The right heart (right atrium and ventricle) pumps deoxygenated blood returning from the systemic veins into the pulmonary circulation at relatively low pressures. The left heart (left atrium and ventricle) receives blood from the lungs and pumps it round the body to the tissues at higher pressures (Fig. 4.2). Atrioventricular valves (tricuspid on the right side, mitral on the left) separate the atria from the ventricles. The pulmonary valve on the right side of the heart and the aortic valve on the left separate the ventricles from the pulmonary and systemic arterial systems, respectively. Cardiac contraction is coordinated by specialised groups of cells. The cells in the sinoatrial node normally act as the cardiac pacemaker. Subsequent spread of impulses through the heart ensures that atrial contraction is complete before ventricular contraction (systole) begins. At the end of systole the ventricles relax and the atrioventricular valves open, allowing them to refill with blood from the atria (diastole).

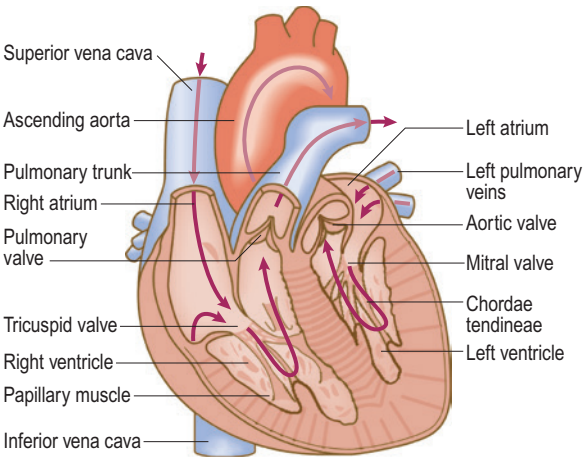


Fig. 4.1 **The heart chambers and valves.**

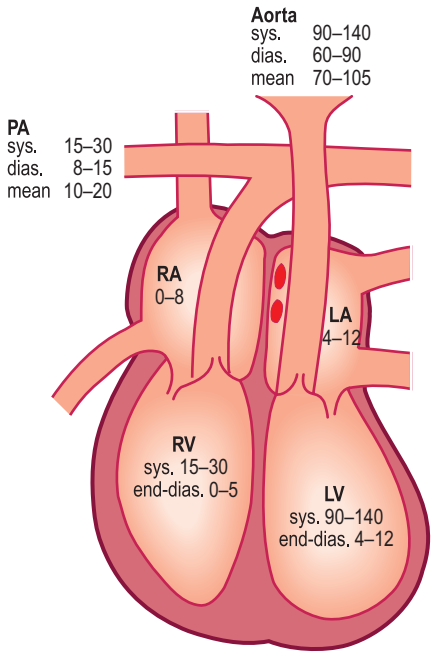


Fig. 4.2 **Normal resting pressures (mmHg) in the heart and great vessels.** *dias.*, diastolic; *LA*, left atrium; *LV*, left ventricle; *PA*, pulmonary artery; *RA*, right atrium; *RV*, right ventricle; *sys.*, systolic.

The history

Common presenting symptoms

Cardiovascular disease may present with a number of diverse symptoms; non-cardiac causes must also be considered (Box 4.1).

Chest pain

Intermittent chest pain

Chest pain due to intermittent myocardial ischaemia (angina pectoris) is typically a dull discomfort, often described as a tight or pressing 'band-like' sensation akin to a heavy weight. It tends to be felt diffusely across the anterior chest and may radiate down one or both arms and into the throat, jaw or teeth. In stable angina (caused by chronic narrowing in one or more coronary arteries), episodes of pain are precipitated by exertion and may occur more readily when walking in cold or windy weather, after

4.1 Common symptoms of heart disease

Symptom	Cardiovascular causes	Other causes
Chest discomfort	Myocardial infarction Angina Pericarditis Aortic dissection	Oesophageal spasm Pneumothorax Musculoskeletal pain
Breathlessness	Heart failure Valvular disease Angina Pulmonary embolism Pulmonary hypertension	Respiratory disease Anaemia Obesity Anxiety
Palpitation	Tachyarrhythmias Ectopic beats	Anxiety Hyperthyroidism Drugs
Syncope/ presyncope	Arrhythmias Postural hypotension Aortic stenosis Hypertrophic cardiomyopathy Atrial myxoma	Simple faints Epilepsy Anxiety
Oedema	Heart failure Constrictive pericarditis Venous stasis Lymphoedema	Nephrotic syndrome Liver disease Drugs Immobility

a large meal or while carrying a heavy load; the pain is promptly relieved by rest and/or sublingual glyceryl nitrate (GTN) spray, and typically lasts for less than 10 minutes. The degree of physical exertion required to precipitate symptoms is a better guide to disease severity than the intensity of discomfort (Box 4.2). In unstable angina (caused by a sudden severe narrowing in a coronary artery), there is usually an abrupt onset or worsening of chest pain episodes that may occur on minimal exertion or at rest. It may be difficult to distinguish between angina and non-cardiac causes of episodic chest pain, such as oesophageal pain or musculoskeletal problems (Box 4.3). The latter may occur at any site over the chest, often vary with posture or specific movements (such as twisting or turning), and may be associated with tenderness to palpation.

Ask about:

- site, onset, severity and character of the pain, and whether the pain radiates anywhere
- associated symptoms such as breathlessness
- aggravating and relieving factors, especially their relationship to exertion
- frequency and duration of symptoms, and any recent change in pattern
- degree of limitation caused by symptoms.

4.2 Canadian Cardiovascular Society: functional classification of stable angina

Grade	Description
1	Ordinary physical activity, such as walking and climbing stairs, does not cause angina. Angina with strenuous, rapid or prolonged exertion at work or during recreation
2	Slight limitation of ordinary activity. Walking or climbing stairs rapidly, walking uphill, walking or climbing stairs after meals, in cold, in wind, or when under emotional stress, or only during the few hours after awakening
3	Marked limitation of ordinary physical activity. Walking 1–2 blocks on the level and climbing less than one flight in normal conditions
4	Inability to carry on any physical activity without discomfort; angina may be present at rest

Acute chest pain

Myocardial infarction causes symptoms that are similar to, but more severe and prolonged than, those of angina pectoris. Associated features include restlessness, breathlessness and a feeling of impending death (*angor animi*). Autonomic stimulation may result in sweating, pallor, nausea and vomiting.

Pericardial pain is typically a constant anterior central chest pain that may radiate to the shoulders. It tends to be sharp or

4.3 Cardiovascular causes of chest pain and their characteristics

	Angina	Myocardial infarction	Aortic dissection	Pericardial pain	Oesophageal pain
Site	Retrosternal	Retrosternal	Interscapular/retrosternal	Retrosternal or left-sided	Retrosternal or epigastric
Onset	Progressive increase in intensity over 1–2 minutes	Rapid over a few minutes	Very sudden	Gradual; postural change may suddenly aggravate	Over 1–2 minutes; can be sudden (spasm)
Character	Constricting, heavy	Constricting, heavy	Tearing or ripping	Sharp, 'stabbing', pleuritic	Gripping, tight or burning
Radiation	Sometimes arm(s), neck, epigastrium	Often to arm(s), neck, jaw, sometimes epigastrium	Back, between shoulders	Left shoulder or back	Often to back, sometimes to arms
Associated features	Breathlessness	Sweating, nausea, vomiting, breathlessness, feeling of impending death (angor animi)	Sweating, syncope, focal neurological signs, signs of limb ischaemia, mesenteric ischaemia	Flu-like prodrome, breathlessness, fever	Heartburn, acid reflux
Timing	Intermittent, with episodes lasting 2–10 minutes	Acute presentation; prolonged duration	Acute presentation; prolonged duration	Acute presentation; variable duration	Intermittent, often at night-time; variable duration
Exacerbating/ relieving factors	Triggered by emotion, exertion, especially if cold, windy Relieved by rest, nitrates	'Stress' and exercise rare triggers, usually spontaneous Not relieved by rest or nitrates	Spontaneous No manœuvres relieve pain	Sitting up/lying down may affect intensity NSAIDs help	Lying flat/some foods may trigger Not relieved by rest; nitrates sometimes relieve
Severity	Mild to moderate	Usually severe	Very severe	Can be severe	Usually mild but oesophageal spasm can mimic myocardial infarction
Cause	Coronary atherosclerosis, aortic stenosis, hypertrophic cardiomyopathy	Plaque rupture and coronary artery occlusion	Thoracic aortic dissection rupture	Pericarditis (usually viral, also post myocardial infarction)	Oesophageal spasm, reflux, hiatus hernia

NSAIDs, *non-steroidal anti-inflammatory drugs.*

stabbing in character, exacerbated by inspiration or lying down, and relieved by sitting forwards. It is caused by inflammation of the pericardium secondary to viral infection, connective tissue disease or myocardial infarction, or after surgery, catheter ablation or radiotherapy.

Aortic dissection (a tear in the intima of the aorta) is usually associated with abrupt onset of very severe, tearing chest pain that can radiate to the back (typically the interscapular region) and may be associated with profound autonomic stimulation. If the tear involves the cranial or upper limb arteries, there may be associated syncope, stroke or upper limb pulse asymmetry. Predisposing factors include hypertension and connective tissue disorders, such as Marfan's syndrome (see Fig. 3.21A–D).

As with intermittent chest pain, explore the characteristics of the pain, and ask specifically about associated symptoms that may guide you to a likely diagnosis, such as interscapular pain, sweating, nausea, vomiting, syncope or neurological features (Box 4.3).

Dyspnoea (breathlessness)

Heart failure is the most common cardiovascular cause of both acute and chronic dyspnoea (Box 4.4). Other cardiovascular causes of acute breathlessness include pulmonary embolism and arrhythmias. Patients with acute heart failure and pulmonary oedema (accumulation of fluid in the alveoli) usually prefer to be upright, while patients with massive pulmonary embolism are often more comfortable lying flat and may faint (syncope) if made to sit upright.

Exertional dyspnoea is the symptomatic hallmark of chronic heart failure. The New York Heart Association grading system is used to assess the degree of symptomatic limitation caused by the exertional breathlessness of heart failure (Box 4.5). Dyspnoea caused by myocardial ischaemia is known as 'angina equivalent'. It may occur instead of, or with, chest discomfort, especially in patients who are elderly or who have diabetes. It has identical precipitants to angina and may be relieved by GTN.

Orthopnoea, dyspnoea on lying flat, may occur in patients with heart failure, where it signifies advanced disease or incipient decompensation. Lying flat increases venous return and in patients with left ventricular impairment may precipitate pulmonary oedema. The severity can be graded by the number of pillows used at night: 'three-pillow orthopnoea', for example. Paroxysmal nocturnal dyspnoea is caused by the same mechanism, resulting in sudden breathlessness that wakes the patient from sleep (Fig. 4.3). Patients may choke or gasp for air, sit on the edge of the bed and open windows in an attempt to relieve their distress. It may be confused with asthma, which can also cause night-time dyspnoea, chest tightness, cough and wheeze, but patients with heart failure may also produce frothy white or blood-stained sputum.

4.4 Some mechanisms and causes of heart failure

Mechanism	Cause
Reduced ventricular contractility (systolic dysfunction)	Myocardial infarction Dilated cardiomyopathy, e.g. genetic, idiopathic, alcohol excess, cytotoxic drugs, peripartum cardiomyopathy Myocarditis
Impaired ventricular filling (diastolic dysfunction)	Left ventricular hypertrophy Constrictive pericarditis Hypertrophic or restrictive cardiomyopathy
Increased metabolic and cardiac demand (rare)	Thyrotoxicosis Arteriovenous fistulae Paget's disease
Valvular or congenital lesions	Mitral and/or aortic valve disease Tricuspid and/or pulmonary valve disease (rare) Ventricular septal defect Patent ductus arteriosus

4.5 New York Heart Association classification of heart failure symptom severity

Class	Description
I	No limitations. Ordinary physical activity does not cause undue fatigue, dyspnoea or palpitation (asymptomatic left ventricular dysfunction)
II	Slight limitation of physical activity. Such patients are comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnoea or angina pectoris (symptomatically 'mild' heart failure)
III	Marked limitation of physical activity. Less than ordinary physical activity will lead to symptoms (symptomatically 'moderate' heart failure)
IV	Symptoms of congestive heart failure are present, even at rest. With any physical activity, increased discomfort is experienced (symptomatically 'severe' heart failure)

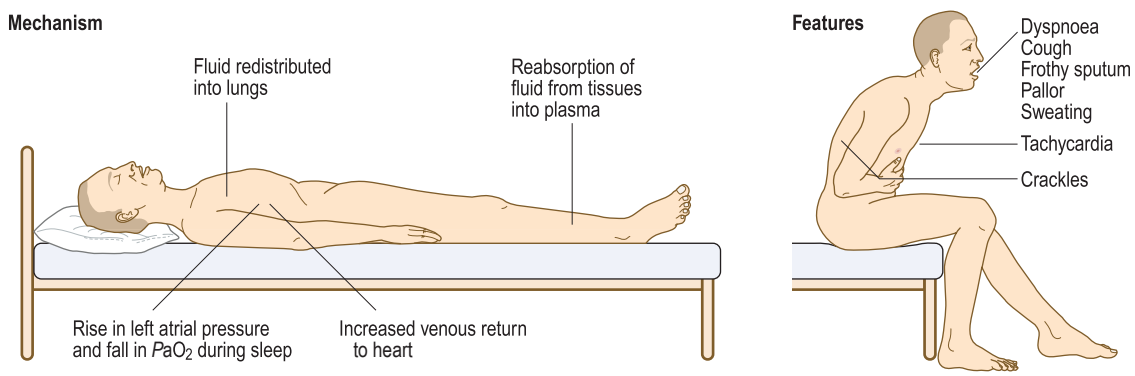


Fig. 4.3 Paroxysmal nocturnal dyspnoea.

In acute dyspnoea, ask about:

- duration of onset
- background symptoms of exertional dyspnoea and usual exercise tolerance
- associated symptoms: chest pain, syncope, palpitation or respiratory symptoms (such as cough, sputum, wheeze or haemoptysis; p. 79).

In patients with chronic symptoms, ask about:

- relationship between symptoms and exertion
- degree of limitation caused by symptoms and their impact on everyday activities
- effect of posture on symptoms and/or episodes of nocturnal breathlessness
- associated symptoms: ankle swelling, cough, wheeze or sputum.

Palpitation

Palpitation is an unexpected or unpleasant awareness of the heart beating in the chest. Detailed history taking can help to distinguish the different types of palpitation (Box 4.6).

Ask about:

- nature of the palpitation: is the heart beat rapid, forceful or irregular? Can the patient tap it out?
- timing of symptoms: speed of onset and offset; frequency and duration of episodes
- precipitants for symptoms or relieving factors
- associated symptoms: presyncope, syncope or chest pain
- history of underlying cardiac disease.

Healthy people are occasionally aware of their heart beating with normal (sinus) rhythm, especially after exercise or in stressful situations such as when waiting for an interview or examination. The sensation is often more common in bed at night and slim people may notice it when lying on their left side.

Ectopic beats (extrasystoles) are a benign cause of palpitation at rest and are abolished by exercise. The premature ectopic beat produces a small stroke volume and an impalpable impulse due to incomplete left ventricular filling. The subsequent compensatory pause leads to ventricular overfilling and a forceful contraction with the next beat. Accordingly, patients often describe 'missed beats', sometimes followed by a particularly strong heart beat ('jolt' or 'thump').

Supraventricular tachycardia produces sudden paroxysms of rapid, regular palpitation that can sometimes be terminated with vagal stimulation using Valsalva breathing manœuvres or carotid sinus pressure. It often affects young patients with no other underlying cardiac disease. Ventricular tachycardia can produce similar symptoms but is more commonly associated with presyncope or syncope, and tends to affect patients with cardiomyopathy or previous myocardial infarction.

High-risk features that increase the likelihood of a life-threatening arrhythmia such as ventricular tachycardia include:

- previous myocardial infarction or cardiac surgery
- associated syncope or severe chest pain
- family history of sudden death
- Wolff–Parkinson–White syndrome
- significant structural heart disease such as hypertrophic cardiomyopathy or aortic stenosis.

Syncope and presyncope

Syncope is a transient loss of consciousness due to transient cerebral hypoperfusion. Causes include postural hypotension, neurocardiogenic syncope, arrhythmias and mechanical obstruction to cardiac output. The same mechanisms may lead to a sensation of lightheadedness and impending loss of consciousness without progressing to actual loss of consciousness (presyncope). The main differential diagnosis of syncope is seizure (p. 122), while lightheadedness and presyncope must be distinguished from dizziness or vertigo due to non-cardiovascular causes (p. 123).

In patients who present with syncope, ask about:

- circumstances of the event and any preceding symptoms: palpitation, chest pain, lightheadedness, nausea, tinnitus, sweating or visual disturbance
- duration of loss of consciousness, appearance of the patient while unconscious and any injuries sustained (a detailed witness history is extremely helpful)
- time to recovery of full consciousness and normal cognition
- current driving status, including occupational driving.

4.6 Descriptions of arrhythmias

	Extrasystoles	Sinus tachycardia	Supraventricular tachycardia	Atrial fibrillation	Ventricular tachycardia
Site	–	–	–	–	–
Onset	Sudden	Gradual	Sudden, with 'jump'	Sudden	Sudden
Character	'Jump', missed beat or flutter	Regular, fast, 'pounding'	Regular, fast	Irregular, usually fast; slower in elderly	Regular, fast
Radiation	–	–	–	–	–
Associated features	Nil	Anxiety	Polyuria, lightheadedness, chest tightness	Polyuria, breathlessness Syncope uncommon	Presyncope, syncope, chest tightness
Timing	Brief	A few minutes	Minutes to hours	Variable	Variable
Exacerbating/ relieving factors	Fatigue, caffeine, alcohol may trigger Often relieved by walking (increases sinus rate)	Exercise or anxiety may trigger	Usually at rest, trivial movements, e.g. bending, may trigger Vagal manœuvres may relieve	Exercise or alcohol may trigger; often spontaneous	Exercise may trigger; often spontaneous
Severity	Mild (usually)	Mild to moderate	Moderate to severe	Very variable, may be asymptomatic	Often severe

4.7 Symptoms related to medication

Symptom	Medication
Angina	Aggravated by thyroxine or drug-induced anaemia, e.g. aspirin or NSAIDs
Dyspnoea	Beta-blockers in patients with asthma Exacerbation of heart failure by beta-blockers, some calcium channel antagonists (verapamil, diltiazem), NSAIDs
Palpitation	Tachycardia and/or arrhythmia from thyroxine, β_2 stimulants, e.g. salbutamol, digoxin toxicity, hypokalaemia from diuretics, tricyclic antidepressants
Syncope/ presyncope	Vasodilators, e.g. nitrates, alpha-blockers, ACE inhibitors and angiotensin II receptor antagonists Bradycardia from rate-limiting agents, e.g. beta-blockers, some calcium channel antagonists (verapamil, diltiazem), digoxin, amiodarone
Oedema	Glucocorticoids, NSAIDs, some calcium channel antagonists, e.g. nifedipine, amlodipine

ACE, *angiotensin-converting enzyme;* NSAIDs, *non-steroidal anti-inflammatory drugs.*

In patients with presyncopal symptoms of lightheadedness or dizziness, ask about:

- exact nature of symptoms and associated features such as palpitation
- precipitants for symptoms, such as postural change, prolonged standing, intense emotion or exertion
- frequency of episodes and impact on lifestyle
- possible contributing medications, such as antihypertensive agents (Box 4.7).

Postural hypotension, a fall of more than 20 mmHg in systolic blood pressure on standing, may lead to syncope or presyncope. It can be caused by hypovolaemia, drugs (Box 4.7) or autonomic neuropathy and is common in the elderly, affecting up to one-third of individuals over 65 years.

In simple faint and other forms of reflex syncope or presyncope, abnormal autonomic reflexes produce a sudden slow heart rate (bradycardia) and/or vasodilatation. These may be triggered in healthy people forced to stand for a long time in a warm environment or subject to painful or emotional stimuli, such as the sight of blood. There is typically a prodrome of lightheadedness, tinnitus, nausea, sweating and facial pallor, and a darkening of vision from the periphery as the retinal blood supply (the most oxygen-sensitive part of the nervous system) is reduced. The person then slides to the floor, losing consciousness. When laid flat to aid cerebral circulation the individual wakes up, often flushing from vasodilatation and nauseated or even vomiting due to vagal overactivity. If the person is held upright by misguided bystanders, continued cerebral hypoperfusion delays recovery and may lead to a seizure and a mistaken diagnosis of epilepsy. In patients with hypersensitive carotid sinus syndrome, pressure over the carotid sinus may lead to reflex bradycardia and syncope.

Arrhythmias can cause syncope or presyncope. The most common cause is bradyarrhythmia caused by sinoatrial disease or atrioventricular block: Stokes–Adams attacks. Rate-limiting drugs are a common cause of bradyarrhythmia. Supraventricular tachyarrhythmias, like atrial fibrillation, rarely cause syncope whereas ventricular tachycardia often causes syncope or presyncope, especially in patients with impaired left ventricular function.

Mechanical obstruction to left ventricular outflow, including severe aortic stenosis and hypertrophic cardiomyopathy, can cause syncope or presyncope, especially on exertion, when cardiac output cannot meet the increased metabolic demand. Massive pulmonary embolism can lead to syncope by obstructing outflow from the right ventricle; associated features are usually apparent and include acute dyspnoea, chest pain and hypoxia. Cardiac tumours, such as atrial myxoma, and thrombosis or failure of prosthetic heart valves are rare causes of syncope.

Oedema

Excess fluid in the interstitial space causes oedema (tissue swelling). It is usually gravity-dependent and so is seen especially around the ankles, or over the sacrum in patients lying in bed. Unilateral lower limb oedema may occur in deep vein thrombosis (p. 70). Heart failure is a common cause of bilateral lower limb oedema but other causes include chronic venous disease, vasodilating calcium channel antagonists (such as amlodipine) and hypoalbuminaemia. An elevated jugular venous pressure strongly suggests a cardiogenic cause of oedema. Enquire about other symptoms of fluid overload, including dyspnoea, orthopnoea and abdominal distension.

Other symptoms of cardiac disease

Infective endocarditis, microbial infection of a heart valve, frequently presents with non-specific symptoms, including weight loss, tiredness, fever and night sweats.

Embolisation of intracardiac thrombus, tumour (such as atrial myxoma) or infective 'vegetations' (Fig. 4.4) may produce symptoms of stroke (p. 123), acute limb ischaemia (p. 65) or acute mesenteric ischaemia (p. 66).

Advanced heart failure may result in either abdominal distension due to ascites, or weight loss and muscle wasting ('cardiac cachexia') due to a prolonged catabolic state.

Past medical history

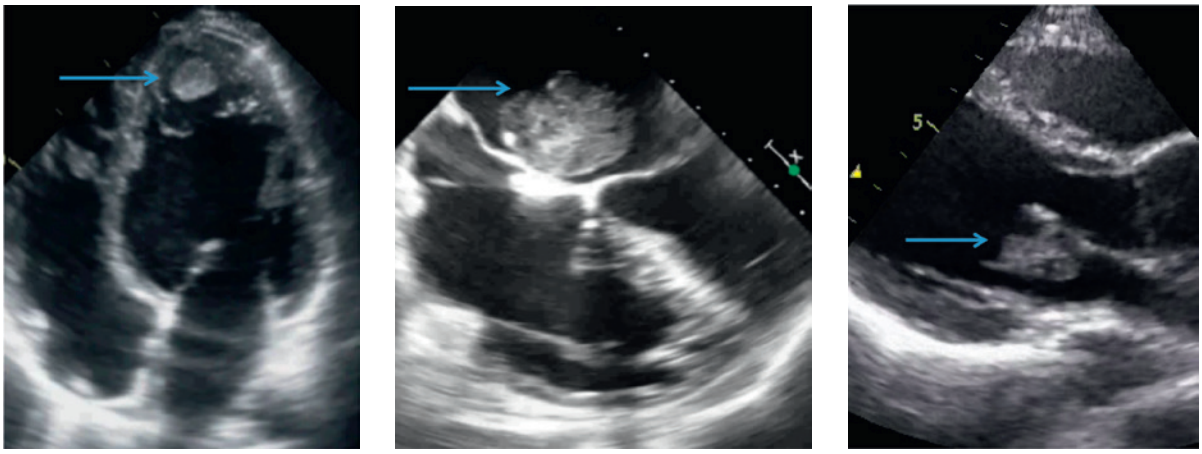
Obtaining a detailed record of any previous cardiac disease, investigations and interventions is essential (Box 4.8). You may need to consult the patient, family members and electronic case records.

Also ask about:

- conditions associated with increased risk of vascular disease such as hypertension, diabetes mellitus and hyperlipidaemia
- rheumatic fever or heart murmurs during childhood
- potential causes of bacteraemia in patients with suspected infective endocarditis, such as skin infection, recent dental work, intravenous drug use or penetrating trauma
- systemic disorders with cardiovascular manifestations such as connective tissue diseases (pericarditis and Raynaud's phenomenon), Marfan's syndrome (aortic dissection) and myotonic dystrophy (atrioventricular block).

Drug history

Drugs may cause or aggravate symptoms such as breathlessness, chest pain, oedema, palpitation or syncope (see Box 4.7). Ask about 'over-the-counter' purchases, such as non-steroidal anti-inflammatory drugs (NSAIDs) and alternative and herbal medicines, as these may have cardiovascular actions.



A B C

Fig. 4.4 Cardiac sources of systemic embolism: echocardiographic images. A A large apical thrombus in the left ventricle *(arrow)*. B An atrial myxoma attached to the interatrial septum *(arrow)*. C A vegetation on the mitral valve *(arrow)* in infective endocarditis. Because all of these lesions are located within the left side of the heart, emboli would flow to the systemic (or coronary) circulation. Conversely, emboli from the right side of the heart would flow to the pulmonary circulation.

4.8 Key elements of the past cardiac history

	Ischaemic heart disease	Heart failure	Valvular disease
Baseline symptoms	Exertional angina? If so, ascertain functional limitation (see Box 4.2)/response to GTN spray	Dyspnoea, fatigue, ankle swelling Record usual functional status (see Box 4.5)	Often asymptomatic Exertional dyspnoea (common), chest pain or syncope
Major events	Previous myocardial infarction/unstable angina	Hospitalisation for decompensated heart failure Ventricular arrhythmias	Infective endocarditis Previous rheumatic fever
Investigations	Coronary angiography (invasive or computed tomography): presence, extent and severity of coronary artery disease Exercise electrocardiogram (or other stress test): evidence of inducible ischaemia? Exercise capacity and symptoms	Echocardiogram (± cardiac magnetic resonance imaging): left ventricular size, wall thickness and systolic function; valvular disease; right ventricular function	Echocardiogram (transthoracic ± transoesophageal): nature and severity of valve lesion; ventricular size and function
Procedures	Percutaneous coronary intervention (angioplasty and stenting) Coronary artery bypass graft surgery	Implantable cardioverter–defibrillator Cardiac resynchronisation therapy	Surgical valve repair or replacement (note whether mechanical or bioprosthetic) Transcatheter valve procedures

GTN, *glyceryl trinitrate*.

Family history

Many cardiac disorders such as cardiomyopathies have a genetic component. Ask about premature coronary artery disease in first-degree relatives (<60 years in a female or <55 years in a male); sudden unexplained death at a young age may raise the possibility of a cardiomyopathy or inherited arrhythmia. Patients with venous thrombosis may have inherited thrombophilia, such as a factor V Leiden mutation. Familial hypercholesterolaemia is associated with premature arterial disease.

Social history

Smoking is the strongest risk factor for coronary and peripheral arterial disease. Take a detailed smoking history (p. 14). Alcohol can induce atrial fibrillation and, in excess, is associated with obesity, hypertension and dilated cardiomyopathy. Recreational drugs such as cocaine and amphetamines can cause arrhythmias, chest pain, occlusive and aneurysmal peripheral arterial disease and even myocardial infarction. Heart disease may have important consequences for employment. Patients with limiting exertional symptoms may struggle to perform jobs that entail a high degree of physical activity. In addition, some diagnoses such as ischaemic heart disease or cardiac arrhythmia may impact on eligibility for certain occupations that have implications for public safety, such as commercial drivers and pilots.

The physical examination

Tailor the sequence and extent of examination to the patient's condition. If you suspect that the person may be unstable, deteriorating or critically unwell (breathless, distressed, cyanosed or obtunded, for example), adopt an ABCDE approach initially (p. 341) and defer detailed examination until stabilised. In stable patients, perform a detailed and comprehensive physical examination.

General examination

Look at the patient's general appearance. Do they look unwell, frightened or distressed? Are there any signs of breathlessness or cyanosis? Is the patient overweight or cachectic? Are there any features of conditions associated with cardiovascular disease such as Marfan's (p. 30), Down's (p. 36) or Turner's syndrome (p. 36), or ankylosing spondylitis (p. 262)?

Conclude by examining the entire skin surface for petechiae, checking the temperature (p. 345) and performing urinalysis (p. 246). Fever is a feature of infective endocarditis and pericarditis, and may occur after myocardial infarction. Urinalysis is necessary to check for haematuria (endocarditis, vasculitis), glucosuria (diabetes) and proteinuria (hypertension and renal disease).

Hands

Examination sequence

- Feel the temperature of the hands and measure capillary refill time (p. 343).
- Examine the hands for tobacco staining (see Fig. 5.8), skin crease pallor (anaemia) or peripheral cyanosis.
- Look at the nails for finger clubbing (p. 24) and for splinter haemorrhages: linear, reddish-brown marks along the axis of the fingernails and toenails (Fig. 4.5B).
- Examine the extensor surface of the hands for tendon xanthomata: hard, slightly yellowish masses over the extensor tendons of the hand from lipid deposits (Fig. 4.6B).
- Examine the palmar aspect of the hands for:
  - Janeway lesions: painless, blanching red macules on the thenar/hypothenar eminences (Fig. 4.5A)
  - Osler's nodes: painful raised erythematous lesions, typically on the pads of the fingers (Fig. 4.5C).

The hands usually feel dry and warm at ambient temperature. Normal capillary refill time is 2 seconds or less. Cool extremities and prolonged capillary refill time signify impaired peripheral

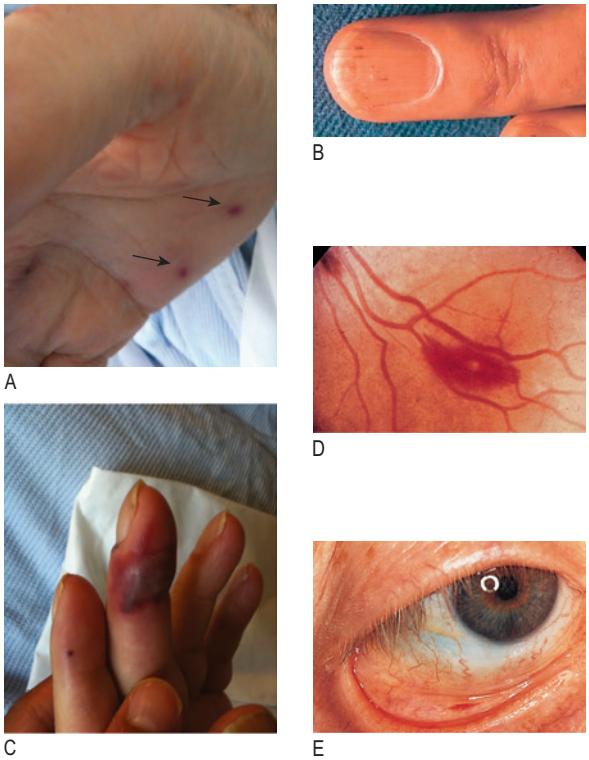


Fig. 4.5 Peripheral signs that may be present in infective endocarditis. A Janeway lesions on the hypothenar eminence *(arrows)*. B Splinter haemorrhages. C Osler's nodes. D Roth's spot on fundoscopy. E Petechial haemorrhages on the conjunctiva. *(B and E) From Walker BR, Colledge NR, Ralston SR, Penman ID, eds. Davidson's Principles and Practice of Medicine. 22nd edn. Edinburgh: Churchill Livingstone; 2014. (D) From Forbes CD, Jackson WF. Color Atlas of Clinical Medicine. 3rd edn. Edinburgh: Mosby; 2003.*

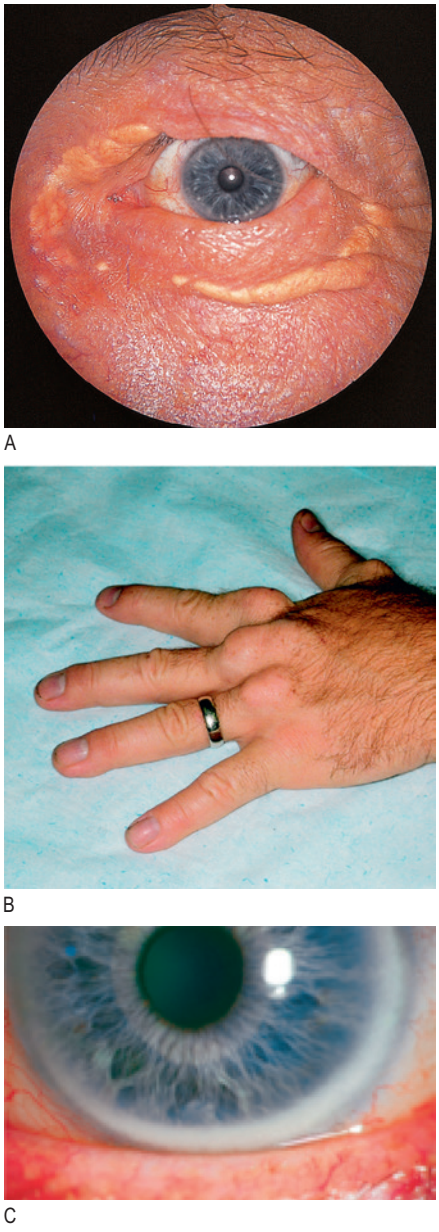


Fig. 4.6 Features of hyperlipidaemia. A Xanthelasmata. B Tendon xanthomata. C Corneal arcus. *(B) From Swartz M. Textbook of Physical Diagnosis. 6th edn. Philadelphia: Saunders; 2009. (C) From Kanski J. Clinical Diagnosis in Ophthalmology. London: Mosby; 2006.*

perfusion, which may occur in shock (p. 343) or chronic conditions associated with a low cardiac output state (as in severe aortic stenosis, mitral stenosis or pulmonary hypertension).

One or two isolated splinter haemorrhages from trauma are common in healthy individuals, especially in manual workers. Splinter haemorrhages (Fig. 4.5B) are found in infective endocarditis and some vasculitic disorders. A petechial rash (caused by vasculitis), most often present on the legs and conjunctivae (Fig. 4.5E), is a transient finding in endocarditis and can be confused with the rash of meningococcal disease. Janeway lesions and Osler's nodes (Fig. 4.5A and C) are features of endocarditis but are rare in the modern era.

Tendon xanthomata (Fig. 4.6) are a sign of familial hypercholesterolaemia, a genetic disorder associated with severe elevations in serum cholesterol and premature coronary artery disease.

Face

Examination sequence

- Look in the mouth for central cyanosis: a purplish blue discoloration of the lips and underside of the tongue (see Fig. 5.11).
- Examine the eyelids for xanthelasmata: soft, yellowish plaques found periorbitally and on the medial aspect of the eyelids (Fig. 4.6A).
- Look at the iris for corneal arcus: a creamy yellow discoloration at the boundary of the iris and cornea (Fig. 4.6C).
- Examine the fundi (p. 164) for features of hypertension (p. 165), diabetes (p. 165) or Roth's spots (flame-shaped retinal haemorrhages with a 'cotton-wool' centre; Fig. 4.5D).

Cardiac causes of central cyanosis include heart failure sufficient to cause pulmonary congestion and oedema impairing gas exchange, or, rarely, congenital heart disease, in which case it is associated with right-to-left shunting and finger clubbing (p. 24).

Xanthelasmata and corneal arcus (Fig. 4.6A and C) are associated with hyperlipidaemia but also occur frequently in normolipidaemic patients. The presence of xanthelasma is an independent risk factor for coronary heart disease and myocardial infarction but corneal arcus has no independent prognostic value.

Arterial pulses

The palpable pulse in an artery reflects the pressure wave generated by the ejection of blood into the circulation from the left ventricle.

When taking a pulse, assess:

- rate: the number of pulses occurring per minute
- rhythm: the pattern or regularity of pulses
- volume: the perceived degree of pulsation
- character: an impression of the pulse waveform or shape.

The rate and rhythm of the pulse are usually determined at the radial artery; use the larger pulses (brachial, carotid or femoral) to assess the pulse volume and character.

Examination sequence

Radial pulse

- Place the pads of your index and middle fingers over the right wrist, just lateral to the flexor carpi radialis tendon (Fig. 4.7A).
- Assess the rhythm of the pulse and count the number over 15 seconds; multiply by 4 to obtain the rate in beats per minute (bpm).
- To detect a collapsing pulse: first, check that the patient has no shoulder or arm pain or restriction on movement; next, feel the pulse with the base of your fingers, then raise the patient's arm vertically above their head (Fig. 4.7B).
- Palpate both radial pulses simultaneously, assessing any delay between the two.

Brachial pulse

- Use your index and middle fingers to palpate the pulse in the antecubital fossa, just medial to the biceps tendon (Fig. 4.7C). Assess the character and volume of the pulse.

Carotid pulse

- Explain what you are going to do.
- With the patient semirecumbent, place the tips of your fingers between the larynx and the anterior border of the sternocleidomastoid muscle (Fig. 4.7D).
- Palpate the pulse gently to avoid a vagal reflex, and never assess both carotids simultaneously.
- Listen for bruits over both carotid arteries, using the diaphragm of your stethoscope in held inspiration.

Rate and rhythm

Resting heart rate is normally 50–95 bpm but should be considered in the clinical context. A pulse rate of 40 bpm can be normal in a fit young adult, whereas a pulse rate of 65 bpm may be abnormally low in acute heart failure. Bradycardia is defined as a pulse rate of <60 bpm; tachycardia is a rate of >100 bpm. The most common causes of bradycardia are medication, athletic conditioning and sinoatrial or atrioventricular node dysfunction.

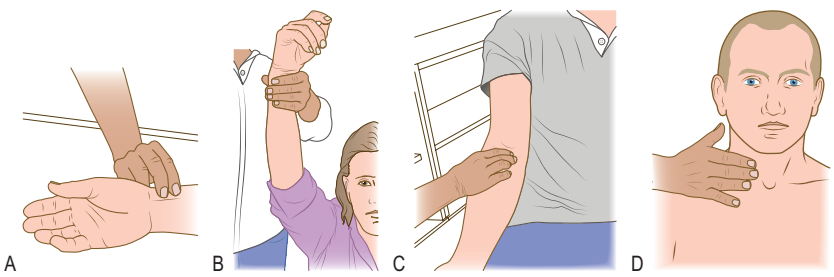


Fig. 4.7 The radial, brachial and carotid pulses. A Locating and palpating the radial pulse. B Feeling for a collapsing radial pulse. C Assessing the brachial pulse. D Locating the right carotid pulse with the fingers.

4.9 Causes of abnormal pulse rate or rhythm

Abnormality	Sinus rhythm	Arrhythmia
Fast rate (tachycardia, >100 bpm)	Exercise Pain Excitement/anxiety Fever Hyperthyroidism Medication: Sympathomimetics, e.g. salbutamol Vasodilators	Atrial fibrillation Atrial flutter Supraventricular tachycardia Ventricular tachycardia
Slow rate (bradycardia, <60 bpm)	Sleep Athletic training Hypothyroidism Medication: Beta-blockers Digoxin Verapamil, diltiazem	Carotid sinus hypersensitivity Sick sinus syndrome Second-degree heart block Complete heart block
Irregular pulse	Sinus arrhythmia Atrial extrasystoles Ventricular extrasystoles	Atrial fibrillation Atrial flutter with variable response Second-degree heart block with variable response

4.10 Haemodynamic effects of respiration

	Inspiration	Expiration
Pulse/heart rate	Accelerates	Slows
Systolic blood pressure	Falls (up to 10 mmHg)	Rises
Jugular venous pressure	Falls	Rises
Second heart sound	Splits	Fuses

4.11 Common causes of atrial fibrillation

- Hypertension
- Heart failure
- Myocardial infarction
- Thyrotoxicosis
- Alcohol-related heart disease
- Mitral valve disease
- Infection, e.g. respiratory, urinary
- Following surgery, especially cardiothoracic surgery

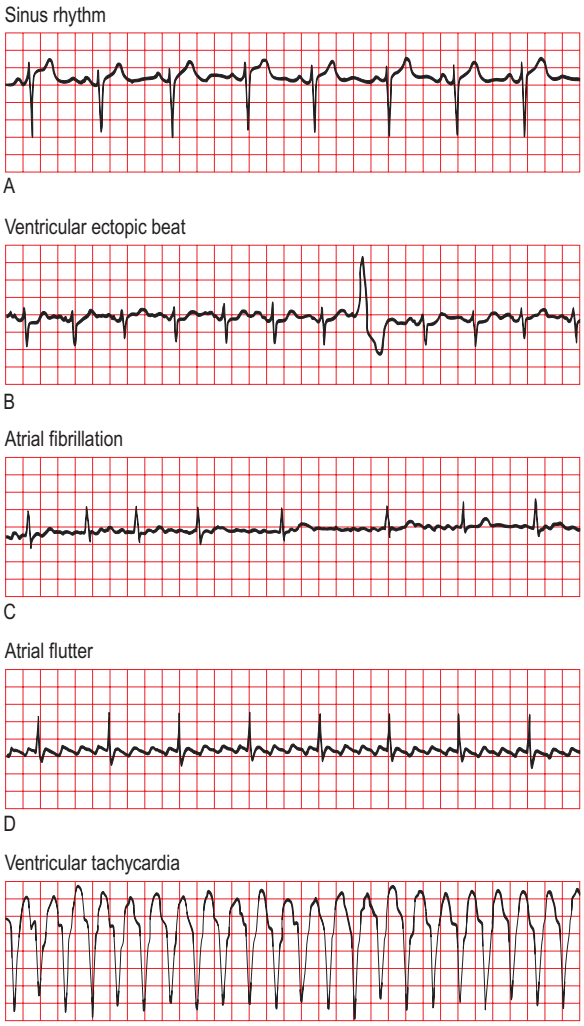


Fig. 4.8 Electrocardiogram rhythm strips. A Sinus rhythm. B Ventricular ectopic beat. C Atrial fibrillation with 'controlled' ventricular response. D Atrial flutter: note the regular 'saw-toothed' atrial flutter waves at about 300/min. E Ventricular tachycardia, with a ventricular rate of about 200/min.

The most common cause of tachycardia is sinus tachycardia (Box 4.9).

The pulse may be regular or irregular (Box 4.9). Sinus rhythm is regular (Fig. 4.8A) but heart rate varies with the respiratory cycle, particularly in children, young adults or athletes (sinus arrhythmia). During inspiration, parasympathetic tone falls and the heart rate increases; on expiration, the heart rate decreases (Box 4.10). With intermittent extrasystoles (Fig. 4.8B) or second-degree atrioventricular block, there may be an underlying regularity to the pulse, interspersed with periods of irregularity (sometimes referred to as 'regularly irregular'). In atrial fibrillation the pulse has no appreciable pattern and is often described as 'irregularly irregular' (Fig. 4.8C and Box 4.11). The rate in atrial fibrillation depends on the number of beats conducted by the atrioventricular node. Untreated, the ventricular rate may be very fast (up to 200 bpm). The variability of the pulse rate (and therefore ventricular filling) explains why the pulse volume varies and there may be a pulse deficit, with some cycles not felt at the radial artery. The pulse deficit can be calculated by counting the radial pulse rate and subtracting this from the apical heart rate, assessed by auscultation.

Volume and character

The ventricles fill during diastole. Longer diastolic intervals are associated with increased stroke volume, which is reflected by increased pulse volume on examination. Abnormalities of pulse volume and character are highly subjective, however, and tend to have poor interobserver agreement.

A large pulse volume is a reflection of a large pulse pressure, which can be physiological or pathological (Box 4.12).

Low pulse volume may result from severe heart failure and conditions associated with inadequate ventricular filling such as hypovolaemia, cardiac tamponade and mitral stenosis. Asymmetric pulses may represent occlusive peripheral arterial disease or

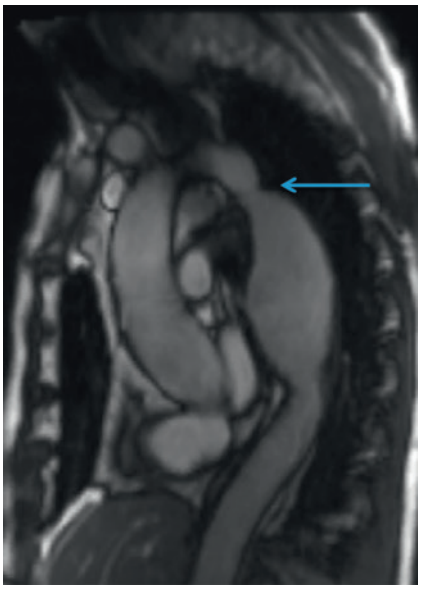


Fig. 4.9 Coarctation of the aorta. Magnetic resonance image showing the typical site of aortic coarctation, just distal to the origin of the left subclavian artery *(arrow)*. This explains why there is synchrony of the radial pulses but radiofemoral delay.

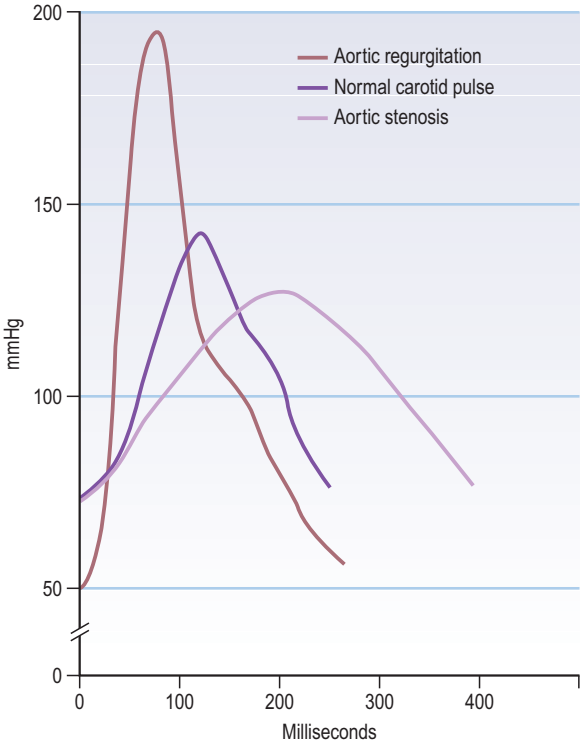


Fig. 4.10 Pulse waveforms.

4.12 Causes of increased pulse volume

Physiological	
• Exercise • Pregnancy • Advanced age	• Increased environmental temperature
Pathological	
• Hypertension • Fever • Thyrotoxicosis • Anaemia	• Aortic regurgitation • Paget's disease of bone • Peripheral atrioventricular shunt

stenosis and, rarely, aortic dissection. Coarctation is a congenital narrowing of the aorta, usually distal to the left subclavian artery (Fig. 4.9); it may produce reduced-volume lower limb pulses, which are also delayed relative to the upper limb pulses (radiofemoral delay). In adults, coarctation usually presents with hypertension and heart failure.

A slow-rising pulse has a gradual upstroke with a reduced peak occurring late in systole, and is a feature of severe aortic stenosis (Fig. 4.10).

A collapsing pulse may occur with severe aortic regurgitation. The peak of the pulse wave arrives early and is followed by a rapid fall in pressure (Fig. 4.10) as blood flows back into the left ventricle, resulting in a wide pulse pressure (systolic – diastolic blood pressure >80 mmHg). This rapid fall imparts the 'collapsing' sensation, and is exaggerated by raising the patient's arm above the level of the heart (see Fig. 4.7B).

Pulsus bisferiens, an increased pulse with a double systolic peak separated by a distinct mid-systolic dip, is classically produced by concomitant aortic stenosis and regurgitation. Pulsus alternans, beat-to-beat variation in pulse volume with a normal rhythm, may occur in advanced heart failure. Both of these signs are rare, however, and of limited relevance in contemporary practice.

Pulsus paradoxus is an exaggeration of the normal variability of pulse volume with breathing. Pulse volume normally increases in expiration and decreases during inspiration due to intrathoracic pressure changes affecting venous return to the heart. This variability is exaggerated when ventricular diastolic filling is impeded by elevated intrapericardial pressure. This is usually due to accumulation of pericardial fluid (cardiac tamponade; Fig. 4.11) but can occur to a lesser extent with pericardial constriction and in acute severe asthma. If suspected, pulsus paradoxus can be confirmed using a blood pressure cuff (see later and Fig. 4.12); a fall of >10 mmHg between the cuff pressure at which Korotkoff sounds appear in expiration only and the cuff pressure at which Korotkoff sounds persist throughout the respiratory cycle is diagnostic.

Blood pressure

Blood pressure (BP) is a measure of the pressure that the circulating blood exerts against the arterial walls. Systolic pressure is the maximal pressure that occurs during ventricular contraction (systole). During ventricular filling (diastole), arterial pressure is maintained at a lower level by the elasticity and compliance of the vessel wall. The lowest value (diastolic pressure) occurs immediately before the next cycle.

BP is usually measured using a sphygmomanometer (Fig. 4.12). In certain situations, such as the intensive care unit, it is measured invasively using an indwelling intra-arterial catheter connected to a pressure sensor.

BP is measured in mmHg and recorded as systolic pressure/diastolic pressure, together with a note of where and how the

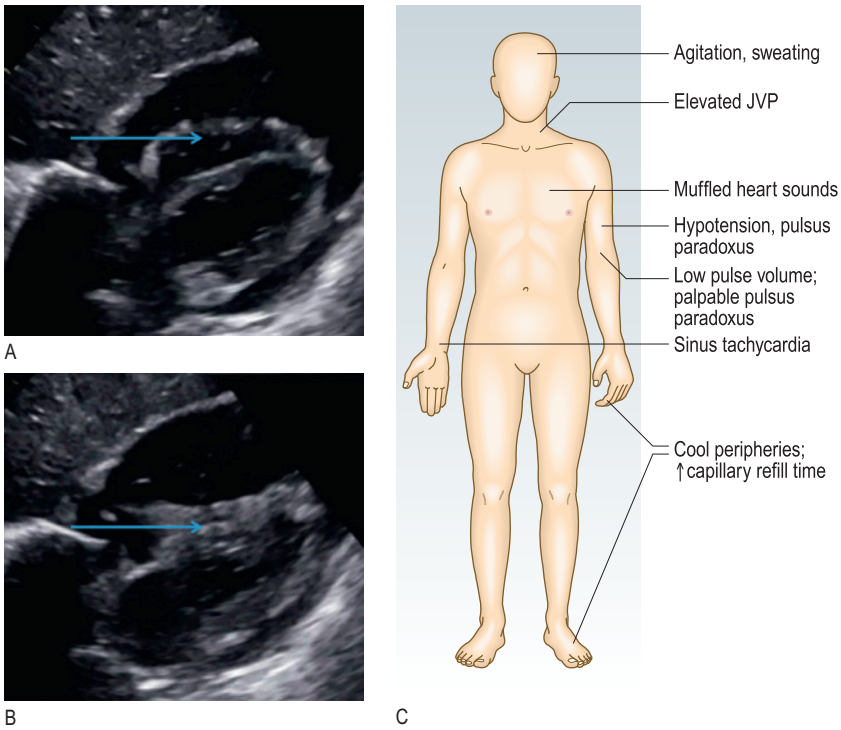


Fig. 4.11 Clinical and echocardiographic features of cardiac tamponade. A and B Echocardiographic images taken from the subcostal position at the onset of systole (A) and in early diastole (B). The right ventricle *(arrows)* is collapsed in the early phase of diastole due to the elevated intrapericardial pressure; this is an important echo finding in tamponade. In both images there is a large pericardial effusion adjacent to the right ventricle. C Clinical features. *JVP,* jugular venous pressure.

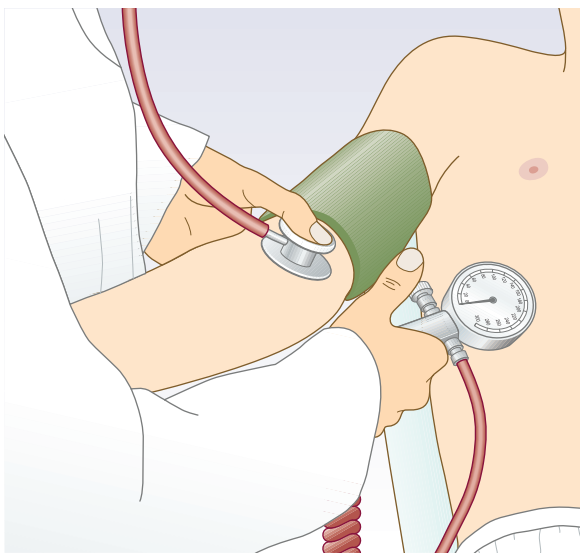


Fig. 4.12 Measuring the blood pressure.

reading was taken: for example, BP 146/92 mmHg, right arm, supine.

BP provides vital information on the haemodynamic condition of acutely ill or injured patients. Over the longer term it is also an important guide to cardiovascular risk. BP constantly varies and rises with stress, excitement and environment. 'White coat hypertension' refers to a transient increase in BP caused by the stress of being in a healthcare setting. Ambulatory BP measurement, using a portable device at intervals during normal daytime activity and at night, is better at determining cardiovascular risk.

4.13 British Hypertension Society classification of blood pressure (BP) levels

BP	Systolic BP (mmHg)	Diastolic BP (mmHg)
Optimal	<120	<80
Normal	<130	<85
High normal	130–139	85–89
Hypertension		
Grade 1 (mild)	140–159	90–99
Grade 2 (moderate)	160–179	100–109
Grade 3 (severe)	>180	>110
Isolated systolic hypertension		
Grade 1	140–159	<90
Grade 2	>160	<90

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Hypertension

Although any threshold for distinguishing abnormal elevation of BP from normal BP is somewhat arbitrary, hypertension is widely defined as a systolic pressure of ≥140 mmHg and/or a diastolic pressure ≥90 mmHg (Box 4.13). Hypertension is associated with significant morbidity and mortality from vascular disease (heart failure, coronary artery disease, cerebrovascular disease and chronic kidney disease). It is almost invariably asymptomatic,

although, rarely and in severe hypertension, headaches and visual disturbances can occur. In most hypertensive patients there is no identifiable cause – so-called 'essential hypertension'. Secondary hypertension is rare, occurring in <1% of the hypertensive population (Box 4.14).

Assess the hypertensive patient for:

- potential underlying causes (Box 4.14)
- end-organ damage:
  - cardiac: heart failure
  - renal: chronic kidney disease, proteinuria
  - eye: hypertensive retinopathy (see Fig. 8.18).

Korotkoff sounds

These sounds are produced when the cuff pressure is between systolic and diastolic because the artery collapses completely and reopens with each heart beat, producing a snapping or knocking sound (Fig. 4.13). The first appearance of sounds (phase 1) during cuff deflation indicates systole. As pressure is gradually reduced, the sounds muffle (phase 4) and then disappear (phase 5).

4.14 Clinical clues to secondary hypertension

Clinical feature	Cause
Widespread vascular disease Renal bruit	Renovascular disease, including renal artery stenosis
Episodes of sweating, headache and palpitation	Phaeochromocytoma
Hypokalaemia	Primary aldosteronism
Cushingoid facies, central obesity, abdominal striae, proximal muscle weakness Chronic glucocorticoid use	Cushing's syndrome
Low-volume femoral pulses with radiofemoral delay	Coarctation of the aorta
Bilateral palpable kidneys	Adult polycystic kidney disease (p. 243)

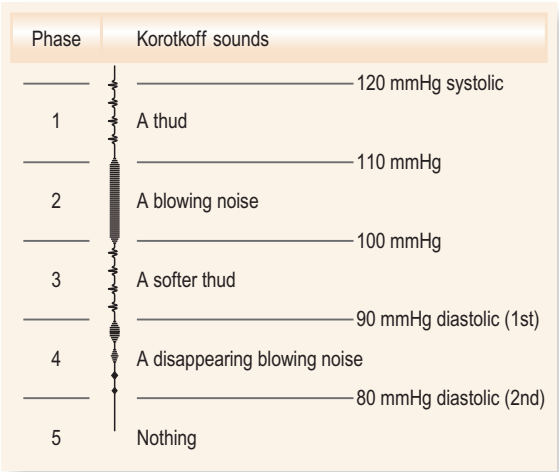


Fig. 4.13 Korotkoff sounds.

Examination sequence

- Rest the patient for 5 minutes.
- Ideally, measure BP in both arms (brachial arteries); the higher of the two is closest to central aortic pressure and should be used to determine treatment.
- With the patient seated or lying down, support their arm comfortably at about heart level, with no tight clothing constricting the upper arm.
- Apply an appropriately sized cuff to the upper arm, with the centre of the bladder over the brachial artery.
- Palpate the brachial pulse.
- Inflate the cuff until the pulse is impalpable. Note the pressure on the manometer; this is a rough estimate of systolic pressure.
- Inflate the cuff another 30 mmHg and listen through the diaphragm of the stethoscope placed over the brachial artery.
- Deflate the cuff slowly (2–3 mmHg/s) until you hear a regular tapping sound (phase 1 Korotkoff sounds). Record the reading to the nearest 2 mmHg. This is the systolic pressure.
- Continue to deflate the cuff slowly until the sounds disappear.
- Record the pressure at which the sounds completely disappear as the diastolic pressure (phase 5). If muffled sounds persist (phase 4) and do not disappear, use the point of muffling as the diastolic pressure.

Common problems in BP measurement

- Different BP in each arm: a difference of >10 mmHg on repeated measurements suggests the presence of aortic or subclavian artery disease. Record the highest pressure and use this to guide management.
- Wrong cuff size: the bladder should be approximately 80% of the length and 40% of the width of the upper arm circumference. A standard adult cuff has a bladder that measures approximately 13×30 cm and suits an arm circumference of 22–26 cm. In obese patients a standard adult cuff will overestimate BP, so use a large adult (bladder 16×38 cm) or thigh cuff (20×42 cm).
- Auscultatory gap: up to 20% of elderly hypertensive patients have Korotkoff sounds that appear at systolic pressure and disappear for an interval between systolic and diastolic pressure. If the first appearance of the sound is missed, the systolic pressure will be recorded at a falsely low level. Avoid this by palpating the systolic pressure first.
- Patient's arm at the wrong level: the patient's elbow should be level with the heart. Hydrostatic pressure causes a change of approximately 5 mmHg in recorded systolic and diastolic BP for a 7 cm change in arm elevation.
- Postural change: the pulse increases by about 11 bpm, systolic BP falls by 3–4 mmHg and diastolic BP rises by 5–6 mmHg when a healthy person stands. The BP stabilises after 1–2 minutes. Check the BP after a patient has been standing for 2 minutes; a drop of >20 mmHg on standing is postural hypotension.
- Atrial fibrillation: in this condition, stroke volume and BP vary from beat to beat, making accurate measurement challenging, so extra care is needed. Reducing cuff

pressure slowly and repeating the measurement more than once will allow an acceptable average value of BP to be obtained.

Jugular venous pressure and waveform

Estimate the jugular venous pressure (JVP) by observing the level of pulsation in the internal jugular vein. The vein runs deep to the sternomastoid muscle and enters the thorax between the sternal and clavicular heads. The normal waveform has two main peaks per cycle, which helps to distinguish it from the carotid arterial pulse (Box 4.15). The external jugular vein is more superficial, prominent and easier to see. It can be kinked or obstructed as it traverses the deep fascia of the neck but, when visible and pulsatile, can be used to estimate the JVP in difficult cases.

The JVP level reflects right atrial pressure (normally <7 mmHg/9 cmH_2O). The sternal angle is approximately 5 cm above the right atrium, so the JVP in health should be ≤4 cm above this angle when the patient lies at 45 degrees (see Fig. 4.15B later). If right atrial pressure is low, the patient may have to lie flat for the JVP to be seen; if high, the patient may need to sit upright (Fig. 4.14).

4.15 Differences between carotid artery and jugular venous pulsation

Carotid	Jugular
Rapid outward movement	Rapid inward movement
One peak per heart beat	Two peaks per heart beat (in sinus rhythm)
Palpable	Impalpable
Pulsation unaffected by pressure at the root of the neck	Pulsation diminished by pressure at the root of the neck
Independent of respiration	Height of pulsation varies with respiration
Independent of the position of the patient	Varies with the position of the patient
Independent of abdominal pressure	Rises with abdominal pressure

Examination sequence

The JVP is usually best seen on the patient's right side.

- Position the patient supine, reclined at 45 degrees, with the head resting on a pillow and turned slightly to the left. The JVP is seen best if the sternocleidomastoid muscles and overlying skin are relaxed, so ensure the head is supported and avoid excessive head turning or elevation of the chin.
- Look across the patient's neck from the right side (Fig. 4.15A). Use oblique lighting if the JVP is difficult to see.
- Identify the jugular vein pulsation behind the sternocleidomastoid muscle (usually just above the clavicle, unless it is elevated).
- If a pulsation is visualised, use the abdominojugular test and/or occlusion to help confirm that it is the JVP.
- The JVP is the vertical height in centimetres between the upper limit of the venous pulsation and the sternal angle (junction of the manubrium and sternum at the level of the second costal cartilages; Fig. 4.15B).
- Identify the timing and waveform of the pulsation and note any abnormality.

It can be difficult to differentiate the jugular venous waveform from arterial pulsation (Box 4.15). If that is the case, the following may help:

- Abdominojugular test: press firmly over the abdomen. This increases venous return to the right side of the heart temporarily and the JVP normally rises.
- Changes with respiration: the JVP normally falls with inspiration due to decreased intrathoracic pressure.
- Waveform: the normal JVP waveform has two distinct peaks per cardiac cycle (Fig. 4.15C):
  - The 'a' wave corresponds to right atrial contraction and occurs just before the first heart sound. In atrial fibrillation the 'a' wave is absent.
  - The 'v' wave is caused by atrial filling during ventricular systole when the tricuspid valve is closed.
  - Rarely, a third peak ('c' wave) may be seen due to closure of the tricuspid valve.
- Occlusion: the JVP waveform is obliterated by gently occluding the vein at the base of the neck with your finger.
- Changes with position: the JVP will vary with the position of the patient (see Fig. 4.14).

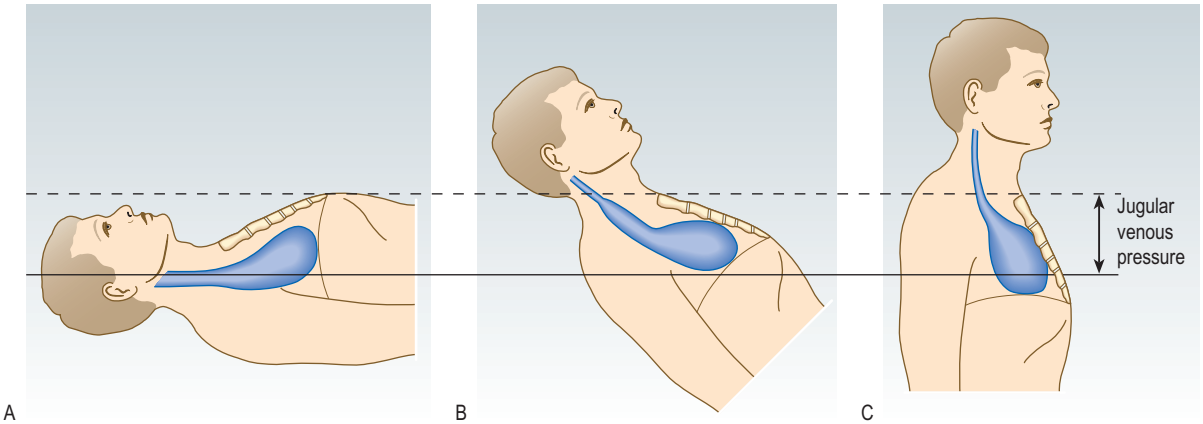


Fig. 4.14 Jugular venous pressure in a healthy subject. A Supine: jugular vein distended, pulsation not visible. B Reclining at 45 degrees: the point of transition between the distended and the collapsed vein can usually be seen to pulsate just above the clavicle. C Upright: the upper part of vein is collapsed and the transition point obscured.

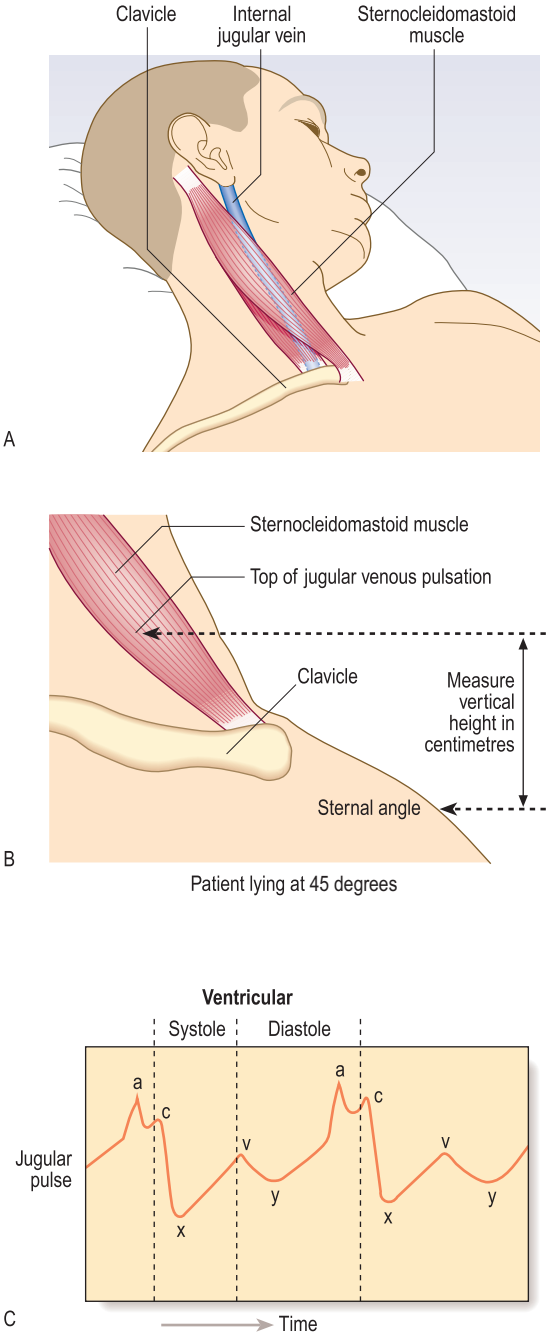


Fig. 4.15 Jugular venous pressure. A Inspecting the jugular venous pressure from the side (the internal jugular vein lies deep to the sternocleidomastoid muscle). B Measuring the height of the jugular venous pressure. C Form of the venous pulse wave tracing from the internal jugular vein: *a,* atrial systole; *c,* closure of the tricuspid valve; *v,* peak pressure in the right atrium immediately prior to opening of the tricuspid valve; *a–x,* descent, due to downward displacement of the tricuspid ring during systole; *v–y,* descent at the commencement of ventricular filling.

The JVP provides a guide to central venous pressure. It is elevated in states of fluid overload, particularly heart failure. Examine the patient for pulmonary oedema or pleural effusions (p. 88), ascites (p. 109) and/or peripheral oedema (p. 245). It is also elevated in any condition that leads to high right ventricular filling pressures, such as pulmonary embolism, chronic pulmonary hypertension, cardiac tamponade (see Fig. 4.11) or pericardial constriction (Box 4.16). Mechanical obstruction of the superior vena cava (most often caused by lung cancer) may cause extreme, non-pulsatile elevation of the JVP. In this case the JVP no longer reflects right atrial pressure and the abdominojugular test will be negative.

Kussmaul's sign is a paradoxical rise of JVP on inspiration that is seen in pericardial constriction, severe right ventricular failure and restrictive cardiomyopathy.

Prominent 'a' waves are caused by delayed or restricted right ventricular filling, as in pulmonary hypertension or tricuspid stenosis.

Cannon waves (giant 'a' waves) occur when the right atrium contracts against a closed tricuspid valve. Irregular cannon waves are seen in complete heart block and are due to atrioventricular dissociation. Regular cannon waves occur during junctional rhythm and with some ventricular and supraventricular tachycardias.

Tricuspid regurgitation results in prominent systolic 'v' waves, which can fuse with the 'c' wave to produce 'cv' waves; there may be an associated pulsatile liver.

4.16 Abnormalities of the jugular venous pulse

Condition	Abnormalities
Heart failure	Elevation, sustained abdominojugular reflux >10 seconds
Pulmonary embolism	Elevation
Pericardial effusion	Elevation, prominent 'y' descent
Pericardial constriction	Elevation, Kussmaul's sign
Superior vena cava obstruction	Elevation, loss of pulsation
Atrial fibrillation	Absent 'a' waves
Tricuspid stenosis	Giant 'a' waves
Tricuspid regurgitation	Giant 'v' or 'cv' waves
Complete heart block	'Cannon' waves

Precordium

The precordium is the anterior chest surface overlying the heart and great vessels (Fig. 4.16).

Learn the surface anatomy and basic physiology of the heart to understand the basis and timing of the heart sounds and murmurs, and why they are heard best in different locations and radiate in a particular direction (Figs 4.16 and 4.17). The optimal sites for auscultation (aortic, pulmonary, apex and left sternal border) do not correspond with the location of cardiac structures but are where the transmitted sounds and murmurs are best heard (Box 4.17). It is important to note that the heart sounds and some murmurs can be heard widely across the precordium, but these sites represent the surface location where the murmur is loudest or easiest to hear.

Inspection

Pectus excavatum (funnel chest; see Fig. 5.6D), a posterior displacement of the lower sternum, and pectus carinatum (pigeon chest; see Fig. 5.6C) may displace the heart and affect palpation and auscultation.

A midline sternotomy scar usually indicates previous valve replacement or coronary artery bypass surgery, in which case

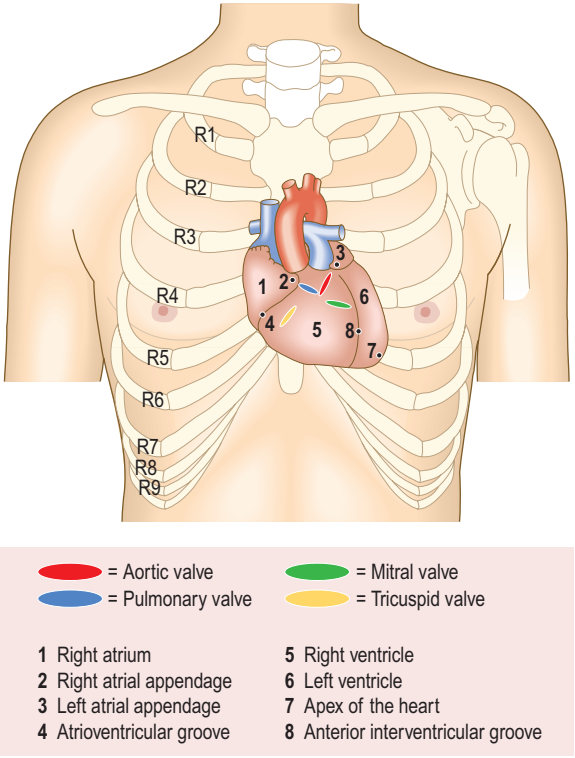


Fig. 4.16 **Surface anatomy of the chambers and valves of the heart.**

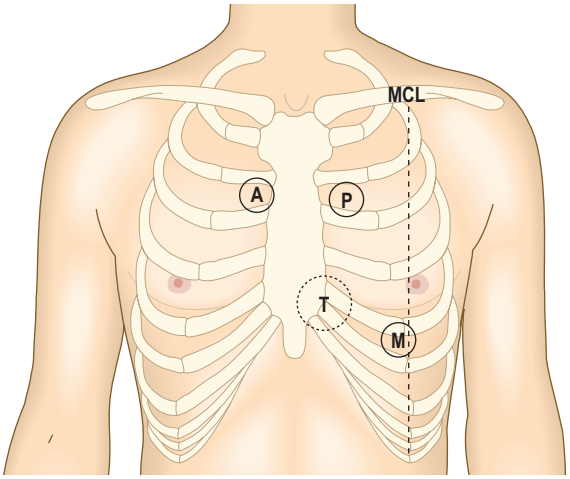


Fig. 4.17 **Sites for auscultation.** Sites at which murmurs from the relevant valves are usually, but not preferentially, heard. *A,* aortic; *M,* mitral; *MCL,* mid-clavicular line; *P,* pulmonary; *T,* tricuspid.

it may be accompanied by saphenous vein or radial artery graft harvest scars. A left submammary scar is usually the result of mitral valvotomy or transapical transcatheter aortic valve implantation. Infraclavicular scars are seen after pacemaker or defibrillator implantation, and the bulge of the device may be obvious.

Palpation

The apex beat may be visible on inspection but is defined as the most lateral and inferior position at which the cardiac impulse can be felt. The cardiac impulse results from the left ventricle moving forwards and striking the chest wall during systole. The apex beat is normally in the fifth left intercostal space at, or medial to, the mid-clavicular line (halfway between the suprasternal notch and the acromioclavicular joint), but may be displaced laterally to the anterior or mid-axillary line, or inferiorly to the sixth or seventh intercostal space when the left ventricle is dilated. A heave is a palpable impulse that noticeably lifts your hand. A thrill is the tactile equivalent of a murmur and is a palpable vibration.

4.17 Cardiac auscultation: the best sites for hearing an abnormality

Site	Sound
Cardiac apex	First heart sound Third and fourth heart sounds Mid-diastolic murmur of mitral stenosis
Lower left sternal border	Early diastolic murmurs of aortic and tricuspid regurgitation Opening snap of mitral stenosis Pansystolic murmur of ventricular septal defect
Upper left sternal border	Second heart sound Pulmonary valve murmurs
Upper right sternal border	Systolic ejection (outflow) murmurs, e.g. aortic stenosis, hypertrophic cardiomyopathy
Left axilla	Radiation of the pansystolic murmur of mitral regurgitation
Below left clavicle	Continuous 'machinery' murmur of a persistent patent ductus arteriosus

Examination sequence

- Explain that you wish to examine the chest and ask the patient to remove all clothing above the waist. Keep a female patient's chest covered with a sheet as far as possible.
- Inspect the precordium with the patient sitting at a 45-degree angle with shoulders horizontal. Look for surgical scars, visible pulsations and chest deformity.
- Place your right hand flat over the precordium to obtain a general impression of the cardiac impulse (Fig. 4.18A).
- Locate the apex beat by laying your fingers on the chest parallel to the rib spaces; if you cannot feel it, ask the patient to roll on to their left side (Fig. 4.18B).
- Assess the character of the apex beat and note its position.
- Apply the heel of your right hand firmly to the left parasternal area and feel for a right ventricle heave. Ask the patient to hold their breath in expiration (Fig. 4.18C).
- Palpate for thrills at the apex and on both sides of the sternum using the flat of your fingers.

A normal apical impulse briefly lifts your fingers and is localised. There should be no parasternal heave or thrill. The apex beat may be impalpable in overweight or muscular people, or in patients with asthma or emphysema because the lungs are hyperinflated. It may be diffusely displaced inferiorly and laterally in left ventricular dilatation, such as after myocardial infarction or in dilated cardiomyopathy, severe aortic regurgitation or decompensated aortic stenosis. In dextrocardia the cardiac apex is palpable on the right side but this condition is uncommon, with a prevalence of 1 : 10,000.

Left ventricular hypertrophy, as in hypertension or severe aortic stenosis, produces a forceful but undisplaced apical impulse. This thrusting apical 'heave' is quite different from the diffuse impulse of left ventricular dilatation. Pulsation over the left parasternal area (right ventricular heave) indicates right ventricular hypertrophy or dilatation, most often accompanying pulmonary hypertension. The 'tapping' apex beat in mitral stenosis represents a palpable first heart sound and is not usually displaced. A double apical impulse is characteristic of hypertrophic cardiomyopathy.

The most common thrill is that of aortic stenosis, which is usually palpable over the upper right sternal border. The thrill caused by a ventricular septal defect is best felt at the left and right sternal edges. Diastolic thrills are very rare.

Auscultation

Correct identification and characterisation of the heart sounds and of any added sounds and/or murmurs require a careful, systematic approach to auscultation.

The diaphragm attenuates all frequencies equally, thus making some low-frequency sounds less audible. Use the diaphragm to identify the first and second heart sounds, and high-pitched sounds such as the early diastolic murmur of aortic regurgitation. Listen with the diaphragm over the whole precordium for a pericardial friction rub.

The bell of the stethoscope transmits all sounds well, but in patients with high-frequency murmurs additional low-frequency sounds may be masked by the high-frequency murmur. The bell is particularly useful at the apex and left sternal edge to listen for the diastolic murmur of mitral stenosis and third and fourth heart sounds.

Examination sequence

Make sure the room is quiet when you auscultate. Your stethoscope should fit comfortably with the earpieces angled slightly forwards. The tubing should be approximately 25 cm long and thick enough to reduce external sound.

- Listen with your stethoscope diaphragm at the:
  - apex
  - lower left sternal border
  - upper right and left sternal borders.
- Listen with your stethoscope bell at the:
  - apex
  - lower left sternal border.
- Listen over the carotid arteries (ejection systolic murmur of aortic stenosis and carotid bruits) and in the left axilla (pansystolic murmur of mitral regurgitation).
- At each site, identify the S_1 and S_2 sounds. Assess their character and intensity; note any splitting of the S_2. Palpate the carotid pulse to time any murmur. The S_1 barely precedes the upstroke of the carotid pulsation, while the S_2 is clearly out of phase with it.
- Concentrate in turn on systole (the interval between S_1 and S_2) and diastole (the interval between S_2 and S_1). Listen for added sounds and then for murmurs. Soft diastolic murmurs are sometimes described as the 'absence of silence'.
- Roll the patient on to their left side. Listen at the apex using light pressure with the bell, to detect the mid-diastolic murmur of mitral stenosis (Fig. 4.19A).
- Ask the patient to sit up and lean forwards, then to breathe out fully and hold their breath (Fig. 4.19B). Listen over the right second intercostal space and over the left sternal edge with the diaphragm for the murmur of aortic regurgitation.
- Note the character and intensity of any murmur heard.
- Develop a routine for auscultation so that you do not overlook subtle abnormalities. Identify and describe the following:
  - the first and second heart sounds (S_1 and S_2)
  - extra heart sounds (S_3 and S_4)
  - additional sounds such as clicks and snaps
  - murmurs in systole and/or diastole (timing, duration, character, pitch, intensity, location and radiation)
  - pericardial rubs.

Heart sounds

Normal heart valves make a sound only when they close. The 'lub-dub' sounds are caused by closure of the atrioventricular (mitral and tricuspid) valves followed by the outlet (aortic and pulmonary) valves.

First heart sound

The first heart sound (S_1), 'lub', is caused by closure of the mitral and tricuspid valves at the onset of ventricular systole. It is best heard at the apex. In mitral stenosis the intensity of S_1 is increased due to elevated left atrial pressure (Box 4.18).

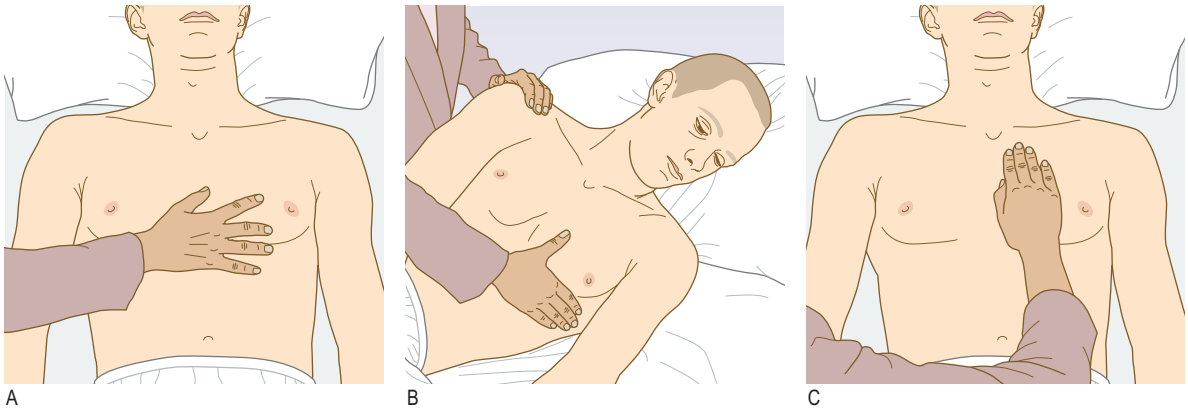
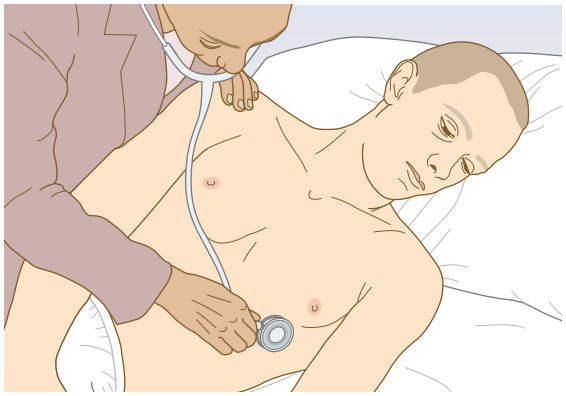
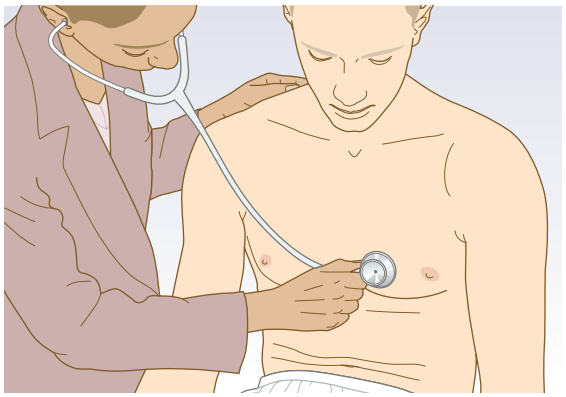


Fig. 4.18 Palpating the heart. A Use your hand to palpate the cardiac impulse. B Localise the apex beat with your finger (if necessary, roll the patient into the left lateral position). C Palpate from apex to sternum for parasternal pulsations.



A



B

Fig. 4.19 Auscultating the heart. A Listen for the murmur of mitral stenosis using the bell lightly applied with the patient in the left lateral position. B Listen for the murmur of aortic regurgitation using the diaphragm with the patient leaning forwards.

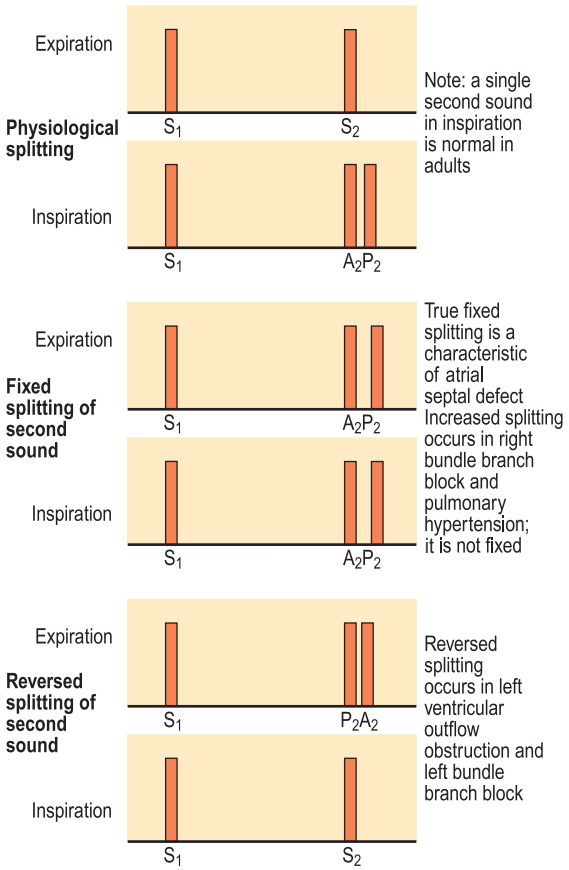


Fig. 4.20 Physiological and pathological splitting of the second heart sound.

4.18 Abnormalities of intensity of the first heart sound

Quiet	
• Low cardiac output • Poor left ventricular function • Rheumatic mitral regurgitation	• Long P–R interval (first-degree heart block)
Loud	
• Increased cardiac output • Large stroke volume	• Mitral stenosis • Short P–R interval • Atrial myxoma (rare)
Variable	
• Atrial fibrillation • Extrasystoles	• Complete heart block

Second heart sound

The second heart sound (S_2), 'dub', is caused by closure of the pulmonary and aortic valves at the end of ventricular systole and is best heard at the left sternal edge. It is louder and higher-pitched than the S_1 'lub', and the aortic component is normally louder than the pulmonary component. Physiological splitting of S_2 occurs because left ventricular contraction slightly precedes that of the right ventricle so that the aortic valve closes before the pulmonary valve. This splitting increases at end-inspiration because increased venous filling of the right ventricle further delays pulmonary valve closure. The separation disappears on expiration (Fig. 4.20). On auscultation, 'lub d-dub' (inspiration) 'lub-dub' (expiration) is heard.

The aortic component of S2 is sometimes quiet or absent in calcific aortic stenosis and reduced in aortic regurgitation (Box 4.19). The aortic component of S_2 is loud in systemic hypertension, and the pulmonary component is increased in pulmonary hypertension.

Wide splitting of S_2, but with normal respiratory variation, occurs in conditions that delay right ventricular emptying, such as right bundle branch block or pulmonary hypertension. Fixed splitting of S_2, with no variation with respiration, is a feature of atrial septal defect (Fig. 4.21). In this condition the right ventricular stroke volume is larger than the left, and the splitting is fixed because the defect equalises the pressure between the two atria throughout the respiratory cycle.

In reversed splitting the two components of S_2 occur together on inspiration and separate on expiration (see Fig. 4.20). This occurs when left ventricular emptying is delayed so that the aortic valve closes after the pulmonary valve. Examples include left bundle branch block and left ventricular outflow obstruction.

Third heart sound

The third heart sound (S_3) is a low-pitched early diastolic sound best heard with the bell at the apex. It coincides with rapid ventricular filling immediately after opening of the atrioventricular

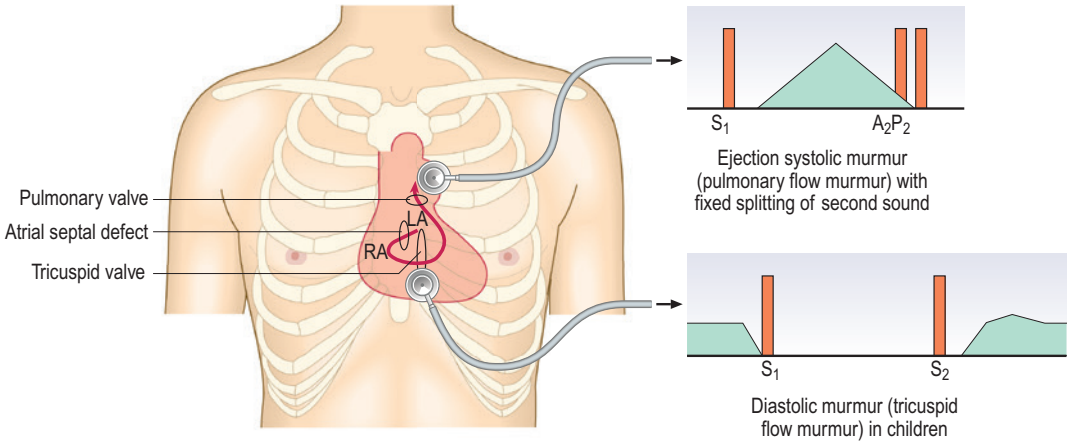


Fig. 4.21 **Atrial septal defect.** *LA,* left atrium; *RA,* right atrium.

4.19 Abnormalities of the second heart sound

Quiet

- Low cardiac output
- Calcific aortic stenosis
- Aortic regurgitation

Loud

- Systemic hypertension (aortic component)
- Pulmonary hypertension (pulmonary component)

Split

Widens in inspiration (enhanced physiological splitting)

- Right bundle branch block
- Pulmonary stenosis
- Pulmonary hypertension
- Ventricular septal defect

Fixed splitting (unaffected by respiration)

- Atrial septal defect

Widens in expiration (reversed splitting)

- Aortic stenosis
- Hypertrophic cardiomyopathy
- Left bundle branch block
- Ventricular pacing

valves and is therefore heard after the second heart sound as 'lub-dub-dum'. It is a normal physiological finding in children, young adults and febrile patients, and during pregnancy, but is usually pathological after the age of 40 years. The most common causes are left ventricular failure, when it is an early sign, and mitral regurgitation, due to volume loading of the ventricle. In heart failure, S_3 occurs with a tachycardia, referred to as a 'gallop' rhythm, and S_1 and S_2 are quiet (lub-da-dub).

Fourth heart sound

The fourth heart sound (S_4) is less common. It is soft and low-pitched, best heard with the bell at the apex. It occurs just before S_1 (da-lub-dub). It is always pathological and is caused by forceful atrial contraction against a non-compliant or stiff ventricle. An S_4 is most often heard with left ventricular hypertrophy (due to hypertension, aortic stenosis or hypertrophic cardiomyopathy). It cannot occur when there is atrial fibrillation.

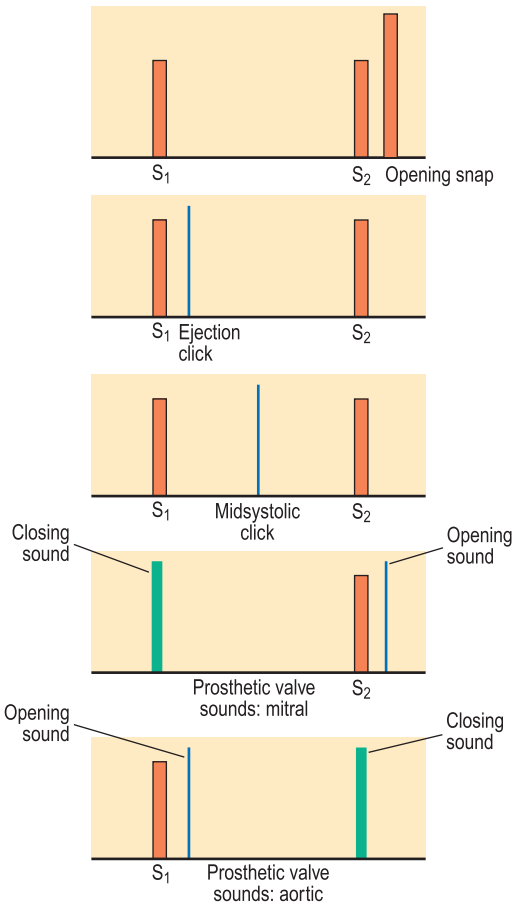


Fig. 4.22 **'Added sounds' on auscultation.**

Added sounds

An opening snap is commonly heard in mitral (rarely, tricuspid) stenosis. It results from sudden opening of a stenosed valve and occurs early in diastole, just after the S_2 (Fig. 4.22A). It is best heard with the diaphragm at the apex.

Ejection clicks are high-pitched sounds best heard with the diaphragm. They occur early in systole just after the S_1, in patients with congenital pulmonary or aortic stenosis (Fig. 4.22B). The

mechanism is similar to that of an opening snap. Ejection clicks do not occur in calcific aortic stenosis because the cusps are rigid.

Mid-systolic clicks are high-pitched and best heard at the apex with the diaphragm. They occur in mitral valve prolapse (Fig. 4.22C) and may be associated with a late systolic murmur.

Mechanical heart valves can make a sound when they close or open. The closure sound is normally louder, especially with modern valves. The sounds are high-pitched, 'metallic' and often palpable, and may even be heard without a stethoscope. A mechanical mitral valve replacement makes a metallic S_1 and a sound like a loud opening snap early in diastole (Fig. 4.22D). Mechanical aortic valves have a loud, metallic S_2 and an opening sound like an ejection click at the start of systole (Fig. 4.22E). They are normally associated with a flow murmur.

Pericardial rub (friction rub) is a coarse scratching sound, often with systolic and diastolic components. It is best heard using the diaphragm with the patient holding their breath in expiration. It may be audible over any part of the precordium but is often localised, varying in intensity over time and with the position of the patient. It is most often heard in acute pericarditis or a few days after an extensive myocardial infarction. A pleuropericardial rub is a similar sound that occurs in time with the cardiac cycle, but is also influenced by respiration and is pleural in origin. Occasionally, a 'crunching' noise can be heard, caused by gas in the pericardium (pneumopericardium).

Murmurs

Heart murmurs are produced by turbulent flow across an abnormal valve, septal defect or outflow obstruction. 'Innocent' murmurs are caused by increased velocity of flow through a normal valve and occur when stroke volume is increased, as in pregnant women, athletes with resting bradycardia or patients with fever.

Examination sequence

Timing

- Identify the first and second heart sounds, S_1 and S_2, respectively. It may help to palpate the patient's carotid pulse while listening to the precordium to determine the onset of ventricular systole. Determine whether the murmur is systolic or diastolic:
  - Systole begins with S_1 (mitral and tricuspid valve closure). This occurs when left and right ventricular pressures exceed the corresponding atrial pressures. For a short period, all four heart valves are closed (pre-ejection period). Ventricular pressures continue to rise until they exceed those of the aorta and pulmonary artery, causing the aortic and pulmonary valves to open. Systole ends with the closure of these valves, producing S_2.
  - Diastole is the interval between S_2 and S_1. Physiologically, it is divided into three phases: early diastole (isovolumic relaxation), the time from the closure of the aortic and pulmonary valves until the opening of the mitral and tricuspid valves; mid-diastole, the early period of ventricular filling when atrial pressures exceed ventricular pressures; and presystole, coinciding with atrial systole.
- Murmurs of aortic and pulmonary regurgitation start in early diastole and extend into mid-diastole. The murmurs of mitral or tricuspid stenosis cannot start before mid-diastole. S_3 occurs in mid-diastole and S_4 in presystole.

Duration

- The murmurs of mitral and tricuspid regurgitation start with S_1, sometimes muffling or obscuring it, and continue throughout systole (pansystolic; Fig. 4.23). The murmur produced by mitral valve prolapse does not begin until the mitral valve leaflet has prolapsed during systole, producing a late systolic murmur (Fig. 4.23). The ejection systolic murmur of aortic or pulmonary stenosis begins after S_1 reaches maximal intensity in mid-systole, then fades, stopping before S_2 (Fig. 4.24).

Character and pitch

- The quality of a murmur is subjective but terms such as harsh, blowing, musical, rumbling and high- or low-pitched can be useful. High-pitched murmurs often correspond to

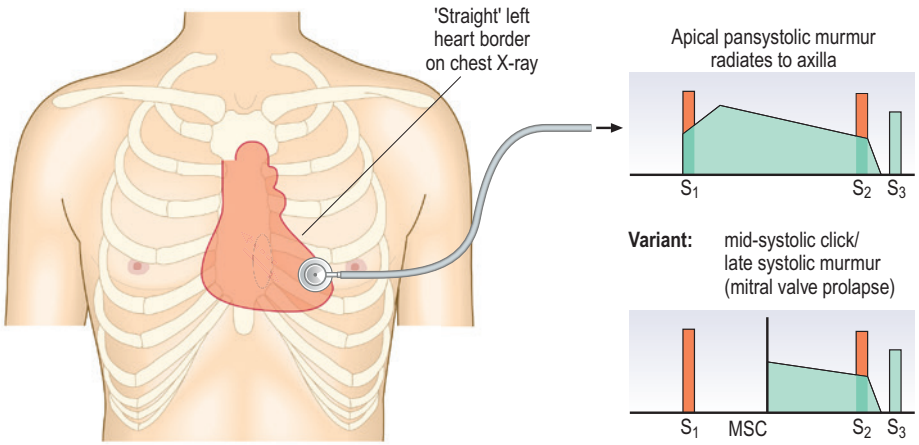


Fig. 4.23 Mitral regurgitation. Mitral regurgitation is caused by dilatation of the left ventricle and failure of leaflets to co-apt. The murmur begins at the moment of valve closure and may obscure the first heart sound. It varies little in intensity throughout systole. In mitral valve prolapse the murmur begins in mid- or late systole and there is often a mid-systolic click (MSC).

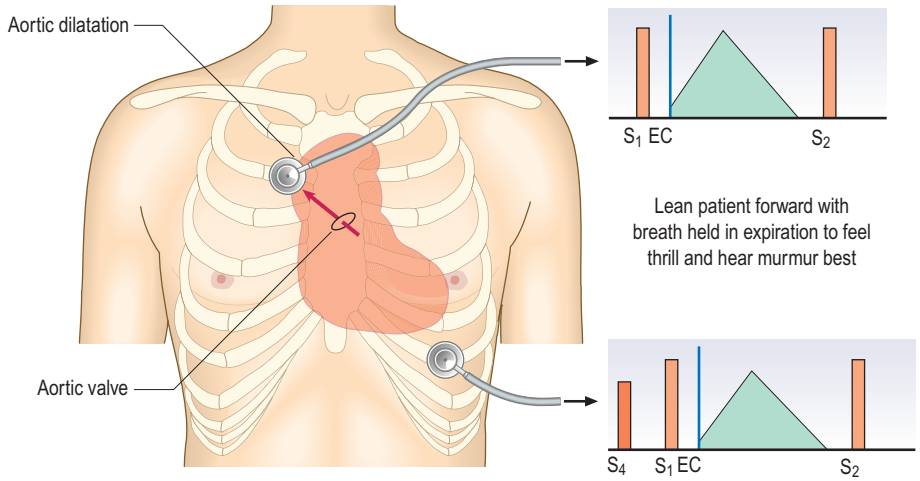


Fig. 4.24 Aortic stenosis. There is a systolic pressure gradient across the stenosed aortic valve. The resultant high-velocity jet impinges on the wall of the aorta *(arrow)*, and is best heard with the diaphragm in the aortic area. Alternatively, the bell may be placed in the suprasternal notch. The ejection systolic murmur follows an ejection click (EC).

4.20 Grades of intensity of murmur

Grade	Description
1	Heard by an expert in optimum conditions
2	Heard by a non-expert in optimum conditions
3	Easily heard; no thrill
4	A loud murmur, with a thrill
5	Very loud, often heard over a wide area, with thrill
6	Extremely loud, heard without a stethoscope

high-pressure gradients, so the diastolic murmur of aortic regurgitation is higher-pitched than that of mitral stenosis.

Intensity

- Describe any murmur according to its grade of intensity (Box 4.20). Diastolic murmurs are rarely louder than grade 3. The intensity of a murmur does not correlate with severity of valve dysfunction; for instance, the murmur of critical aortic stenosis can be quiet and occasionally inaudible. Changes in intensity with time are important, as they can denote progression of a valve lesion. Rapidly changing murmurs can occur with infective endocarditis because of valve destruction.

Location

- Record the site(s) where you hear the murmur best. This helps to differentiate diastolic murmurs (mitral stenosis at the apex, aortic regurgitation at the left sternal edge) but is less helpful with systolic murmurs, which are often audible across the precordium (see Fig. 4.17).

Radiation

- Murmurs radiate in the direction of the blood flow to specific sites outside the precordium. Differentiate radiation from location. The pansystolic murmur of mitral regurgitation radiates towards the left axilla, the murmur of ventricular septal defect towards the right sternal edge, and that of aortic stenosis to the suprasternal notch and the carotid arteries.

4.21 Causes of systolic murmurs

Ejection systolic murmurs

- Increased flow through normal valves:
  - Severe anaemia, fever, athletes (bradycardia → large stroke volume), pregnancy
  - Atrial septal defect (pulmonary flow murmur)
  - Other causes of flow murmurs (increased stroke volume in aortic regurgitation)
- Normal or reduced flow though a stenotic valve:
  - Aortic stenosis
  - Pulmonary stenosis
- Subvalvular obstruction:
  - Hypertrophic obstructive cardiomyopathy

Pansystolic murmurs

- Mitral regurgitation
- Tricuspid regurgitation
- Ventricular septal defect
- Leaking mitral or tricuspid prosthesis

Late systolic murmurs

- Mitral valve prolapse

Systolic murmurs

Ejection systolic murmurs are caused by increased flow through a normal valve (flow or innocent murmur), or by turbulent flow through an abnormal valve, as in aortic or pulmonary stenosis (Box 4.21). An ejection systolic murmur is also a feature of hypertrophic obstructive cardiomyopathy, where it is accentuated by exercise or during the strain phase of the Valsalva manœuvre. An atrial septal defect is characterised by a pulmonary flow murmur during systole.

The murmur of aortic stenosis is often audible all over the precordium (see Fig. 4.24). It is harsh, high-pitched and musical, and radiates to the upper right sternal edge, suprasternal notch and carotid arteries. It is usually loud and there may be a thrill.

Pansystolic murmurs are most commonly caused by mitral regurgitation. The murmur is often loud and blowing in character, and is best heard at the apex and radiating to the axilla. With mitral

valve prolapse, regurgitation begins in mid-systole, producing a late systolic murmur (see Fig. 4.23). The murmur of tricuspid regurgitation is heard at the lower left sternal edge; if significant, it is associated with a prominent 'v' wave in the JVP and a pulsatile liver.

Ventricular septal defects also cause a pansystolic murmur. Small congenital defects produce a loud murmur audible at the left sternal border, radiating to the right sternal border and often associated with a thrill. Rupture of the interventricular septum can complicate myocardial infarction, producing a harsh pansystolic murmur. The differential diagnosis of a murmur heard after myocardial infarction includes acute mitral regurgitation due to papillary muscle rupture, functional mitral regurgitation caused by left ventricular dilatation, and a pericardial rub.

Diastolic murmurs

Early diastolic murmurs usually last throughout diastole but are loudest in early diastole, so the term 'early diastolic murmur' is misleading. The murmur is typically caused by aortic regurgitation (Fig. 4.25), and is best heard at the left sternal edge with the patient leaning forwards in held expiration. The duration of the aortic regurgitation murmur is inversely proportional to severity. Since the regurgitant blood must be ejected during the subsequent systolic period, significant aortic regurgitation increases stroke volume is usually associated with a systolic flow murmur.

Pulmonary regurgitation is very uncommon. It may be caused by pulmonary artery dilatation in pulmonary hypertension (Graham Steell murmur) or a congenital defect of the pulmonary valve.

Mid-diastolic murmurs are usually caused by mitral stenosis. This is a low-pitched, rumbling sound that may follow an opening snap (Fig. 4.26). It is best heard with the bell at the apex and the patient positioned on their left side. The murmur is accentuated by exercise. The cadence sounds like 'lup-ta-ta-rru'; 'lup' is the loud S_1, 'ta-ta' the S_2 and opening snap, and 'rru' the mid-diastolic murmur. If the patient is in sinus rhythm, left atrial contraction increases the blood flow across the stenosed valve, leading to presystolic accentuation of the murmur. The murmur of tricuspid stenosis is similar but rare.

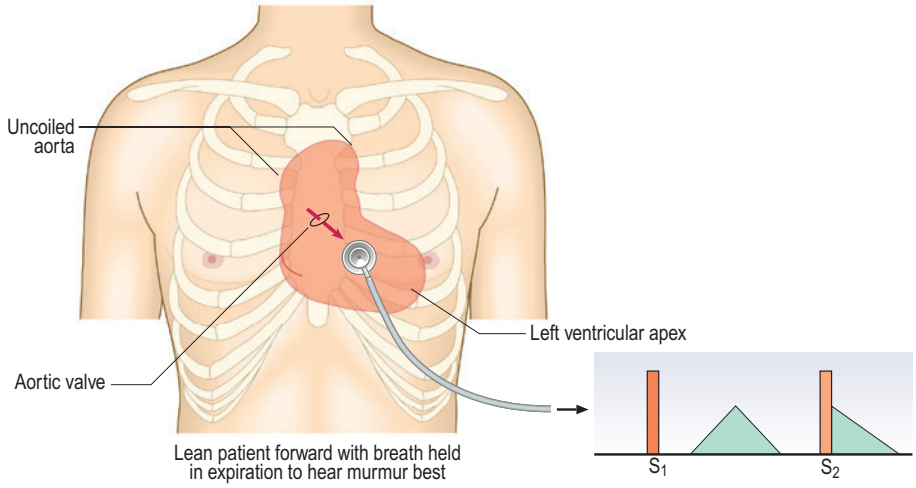


Fig. 4.25 Aortic regurgitation. The pulse pressure is usually increased. The jet from the aortic valve is directed inferiorly towards the left ventricular outflow tract *(arrow)* during diastole, producing an early diastolic murmur, best heard with the diaphragm during held expiration with the patient leaning forward. An associated systolic murmur is common because of the increased flow through the aortic valve in systole.

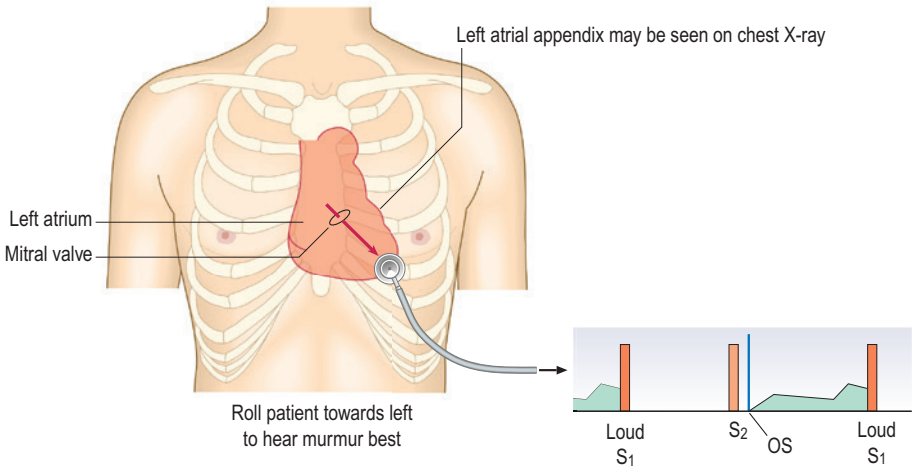


Fig. 4.26 Mitral stenosis. There is a pressure gradient across the mitral valve, giving rise to a low-pitched mid-diastolic murmur that is heard best with the bell at the apex and the patient rolled on to their left-hand side. The jet through the stenosed valve *(arrow)* strikes the endocardium at the cardiac apex. Occasionally, an opening snap (OS) can arise due to the sharp movement of the tethered anterior cusp of the mitral valve at the time when the flow commences.

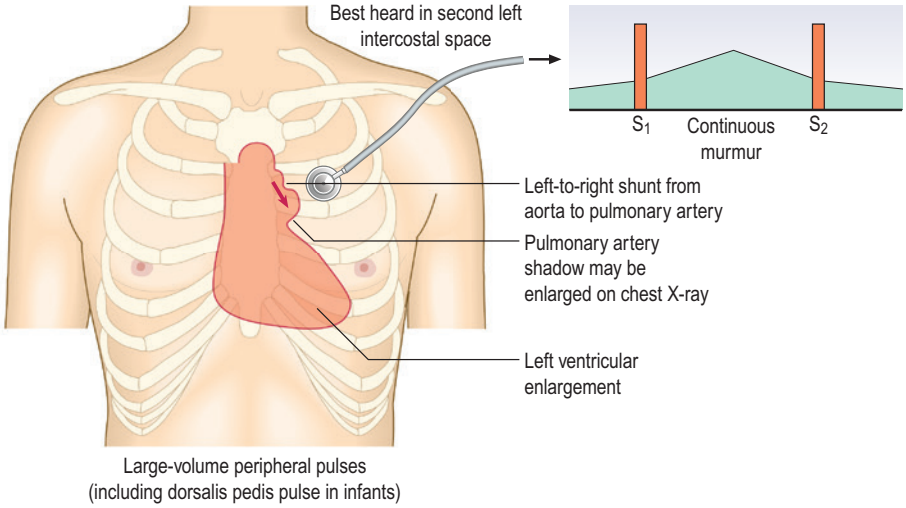


Fig. 4.27 Persistent patent ductus arteriosus. A continuous 'machinery' murmur is heard because aortic pressure always exceeds pulmonary arterial pressure, resulting in continuous ductal flow. The pressure difference is greatest in systole, producing a louder systolic component to the murmur.

An Austin Flint murmur is a mid-diastolic murmur that accompanies aortic regurgitation. It is caused by the regurgitant jet striking the anterior leaflet of the mitral valve, restricting inflow to the left ventricle.

Continuous murmurs

Continuous murmurs are rare in adults. The most common cause is a patent ductus arteriosus. In the fetus this connects the upper descending aorta and pulmonary artery, and normally closes just after birth. The murmur is best heard at the upper left sternal border and radiates over the left scapula. Its continuous character is 'machinery-like'; as aortic pressure always exceeds pulmonary pressure, there is continuous ductal flow, with the greatest pressure difference in systole, resulting in a louder systolic component (Fig. 4.27).

Interpretation of the findings

Auscultation remains an important clinical skill, despite the availability of echocardiography. You must be able to detect abnormal signs to prompt appropriate investigation. Auscultatory signs, such as S_3 or S_4 and pericardial friction rubs, have no direct equivalent on echocardiography but do have diagnostic utility. In addition, some patients, especially those with rheumatic heart disease, have multiple heart valve defects, and the interpretation of more subtle physical signs may be important. For example, a patient with mixed mitral stenosis and regurgitation will probably have dominant stenosis if the S_1 is loud but dominant regurgitation if there is an S_3.

Investigations

Haematology and clinical chemistry

As anaemia can unmask angina or exacerbate heart failure, a full blood count is useful and helps guide the safe use of antiplatelet therapies and anticoagulants. Measurement of the erythrocyte sedimentation rate and serology are indicated if connective tissue disease is suspected. Urea and electrolytes are measured and liver function tests performed prior to starting therapies that may impact on renal function or cause hepatotoxicity. Blood glucose and a lipid profile help identify patients with diabetes mellitus and assess cardiovascular risk. In patients with acute chest pain, cardiac troponin is measured to determine whether there is myocardial injury or infarction.

Electrocardiography

In performing standard 12-lead electrocardiography (ECG; Fig. 4.28), the patient must be resting supine and relaxed to avoid muscle tremor. Good contact between the electrode and skin is important and it may be necessary to shave the chest. The electrodes must be positioned correctly to obtain recordings made from the six precordial electrodes (V_1–V_6) and six recordings from the limb electrodes (left arm, right arm and left leg). The right leg electrode is used as a reference. Confirm that the electrocardiograph is calibrated using a 1 mV signal prior to recording.

Ambulatory ECG monitoring

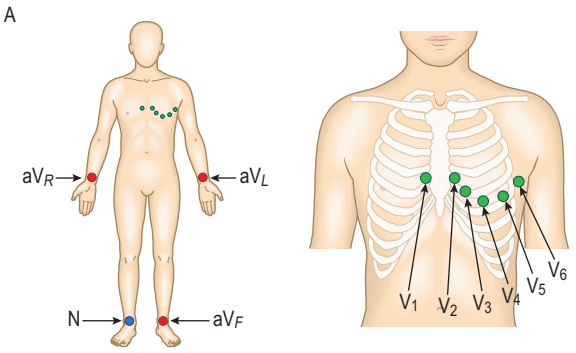
Continuous ECG recording over 24–48 hours can be used to identify symptomatic or asymptomatic rhythm disturbances in patients with palpitation or syncope. If symptoms are less frequent, it may be necessary to use patient-activated recorders that record the heart rhythm only when the patient is symptomatic; the device is activated by the patient (Fig. 4.29).

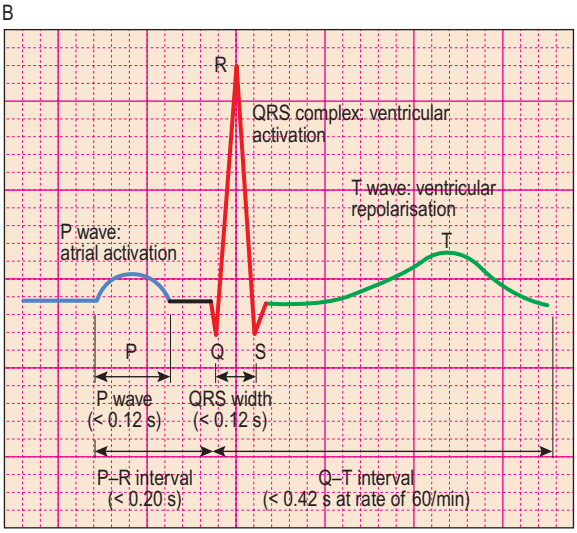
Exercise ECG

An exercise ECG is useful in the diagnosis and functional assessment of patients with suspected coronary artery disease. Down-sloping ST segment depression, particularly when it occurs during minor exertion, or ST segment elevation is of prognostic significance and helps inform the need for invasive investigation with coronary angiography.

Ambulatory blood pressure monitoring

A portable device can be worn by the patient at home that takes at least two BP measurements per hour. It is used to confirm the diagnosis of hypertension and provide a more reliable assessment of response to treatment.





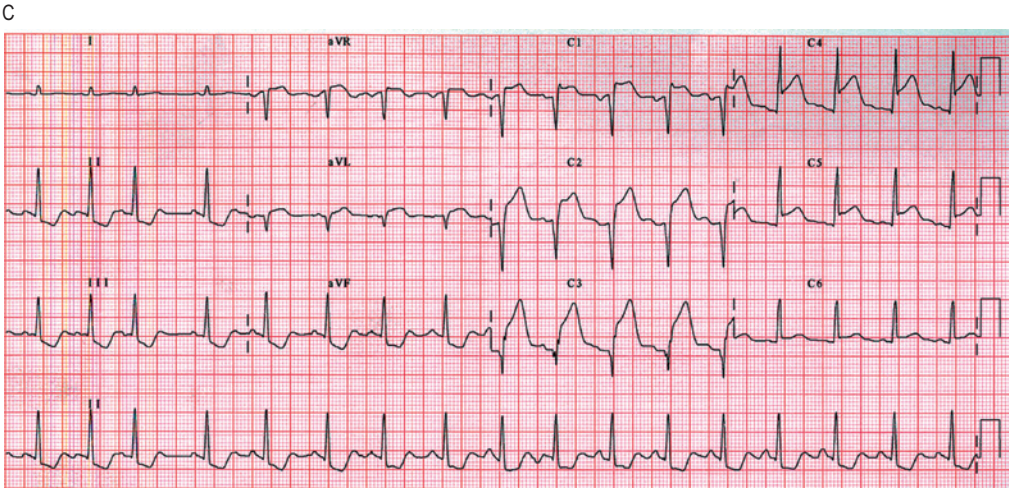


Fig. 4.28 Electrocardiography (ECG). A 12-lead ECG lead placement. B Normal PQRST complex. C Acute anterior myocardial infarction. Note the ST elevation in leads V_1–V_6 and aV_L, and 'reciprocal' ST depression in leads II, III and aV_F.

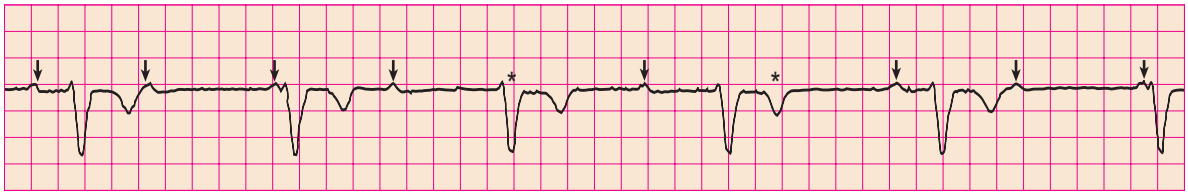


Fig. 4.29 Printout from a 24-hour ambulatory electrocardiogram recording, showing complete heart block. Arrows indicate visible P waves. At times, these are masked by the QRS complex or T wave (*).

Chest X-ray

The maximum width of the heart divided by the maximum width of the thorax on a posteroanterior chest X-ray (the cardiothoracic ratio) should normally be <0.5. An increased cardiothoracic ratio is common in valvular heart disease and heart failure. In heart failure this is often accompanied by distension of the upper lobe pulmonary veins, diffuse shadowing within the lungs due to pulmonary oedema, and Kerley B lines (horizontal, engorged lymphatics at the periphery of the lower lobes; Fig. 4.30A). A widened mediastinum may indicate a thoracic aneurysm.

Echocardiography

Echocardiography uses high-frequency sound waves to evaluate valve abnormalities, ventricular function and blood flow (Doppler echocardiography). Most scans are performed through the anterior chest wall (transthoracic; Fig. 4.30B). Transoesophageal echocardiography requires sedation but gives high resolution of posterior structures such as the left atrium, mitral valve and descending aorta, and is useful in detecting valvular vegetations in infective endocarditis.

Radionuclide studies

Technetium-99 is injected intravenously and detected using a gamma camera to assess left ventricular function. Thallium and sestamibi are taken up by myocardial cells and indicate myocardial perfusion at rest and exercise.

Cardiac catheterisation

A fine catheter is introduced under local anaesthetic via a peripheral artery (usually the radial or femoral) and advanced to the heart under X-ray guidance. Although measurements of intracardiac pressures, and therefore estimates of valvular and cardiac function, are possible, the primary application of this technique is imaging of the coronary circulation using contrast

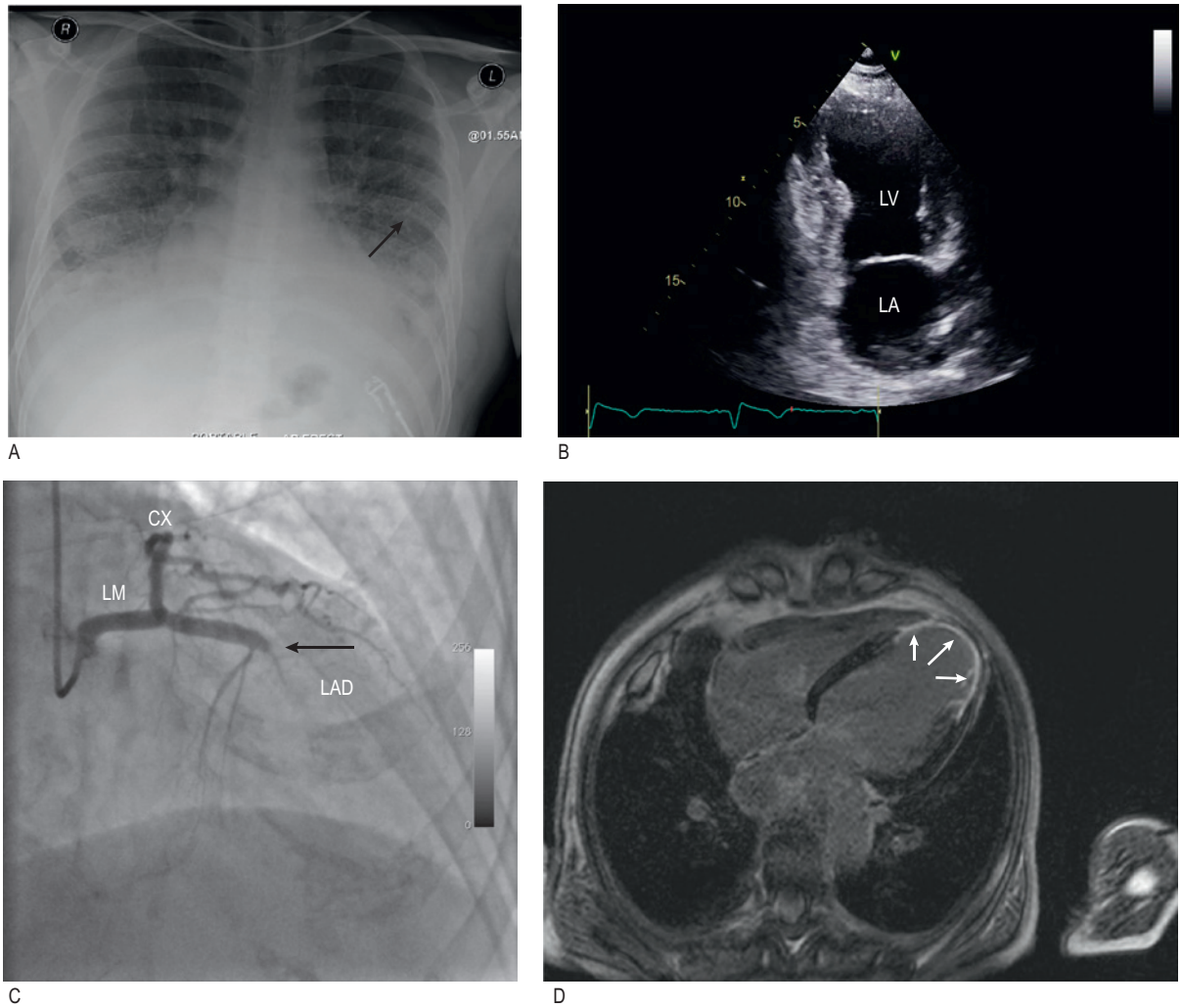


Fig. 4.30 Cardiovascular imaging. A Chest X-ray in heart failure. This shows cardiomegaly with patchy alveolar shadowing of pulmonary oedema and Kerley B lines (engorged lymphatics, *arrow*) at the periphery of both lungs. B Transthoracic echocardiogram in an apical two-chamber view, showing thinning of the left ventricular apex. This is the site of a recent anterior myocardial infarct. *LA,* left atrium; *LV,* left ventricle. C Coronary angiography. The *arrow* indicates an abrupt occlusion of the proximal left anterior descending artery. *CX,* circumflex; *LAD,* left anterior descending; *LM,* left main. D Cardiac magnetic resonance imaging. Gadolinium enhancement image demonstrates regional uptake of gadolinium *(white arrows)* consistent with myocardial fibrosis in the territory of the LAD.

medium. This is performed to inform and guide revascularisation, either by coronary angioplasty or bypass grafting (Fig. 4.30C).

Computed tomography and magnetic resonance imaging

Computed tomography (CT), with its superior temporal resolution of the coronary arteries, is particularly useful for investigating symptomatic patients at low to intermediate risk of coronary artery disease. It can reduce the need for invasive investigation in patients with a low probability of occlusive coronary disease who require valve surgery. Magnetic resonance imaging (MRI) provides superior spatial resolution (Fig. 4.30D) and is the imaging modality of choice for investigating the aetiology of heart muscle diseases (cardiomyopathy).

PERIPHERAL ARTERIAL SYSTEM

Anatomy and physiology

See Fig. 4.31.

The history

Common presenting symptoms

Leg pain

Asymptomatic ischaemia

Approximately 20% of people over 60 years of age in developed countries have peripheral arterial disease (PAD) but only one-quarter of these are symptomatic. The underlying pathology is usually atherosclerosis affecting large and medium-sized vessels.

Haemodynamically significant lower limb ischaemia is defined as an ankle-to-brachial pressure index of <0.9 at rest (p. 69).

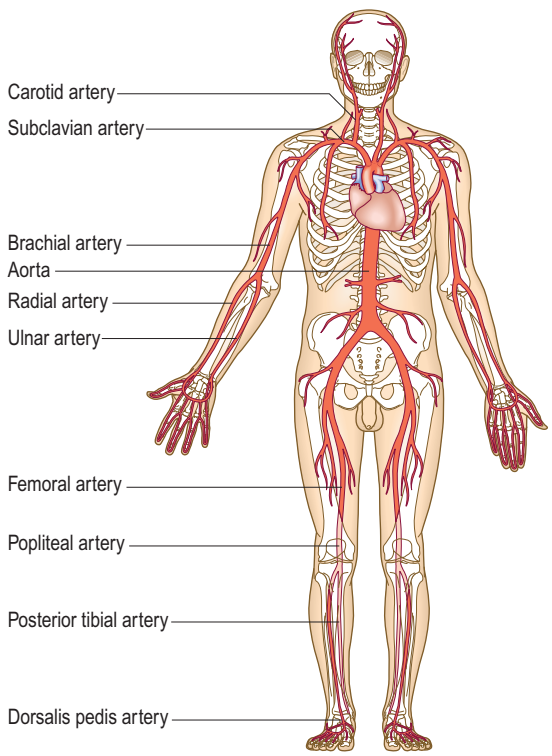


Fig. 4.31 The arterial system.

Most of these patients are asymptomatic, either because they choose not to walk very far or because their exercise tolerance is limited by other comorbidities. Although asymptomatic, these patients have extensive atherosclerosis, putting them at high risk of major cardiovascular events, and should be treated with best medical therapy to reduce their mortality rate. Furthermore, PAD may affect the medical and surgical treatment for other conditions. For example, wound healing may be impaired in the lower limb.

PAD affects the legs eight times more commonly than the arms. This is partly because the lower limb arteries are more frequently affected by atherosclerosis, but also because the arterial supply to the legs is less well developed in relation to the muscle mass.

The Fontaine classification describes the progression of symptoms that occurs as the atherosclerotic burden increases and the blood supply to the limb diminishes (Box 4.22).

Intermittent claudication

Intermittent claudication is pain felt in the legs on walking due to arterial insufficiency and is the most common symptom of PAD. Patients describe tightness or 'cramp-like' pain that develops after a relatively constant distance; the distance is often shorter if walking uphill, in the cold and after meals. The pain disappears completely within a few minutes of rest but recurs on walking. The 'claudication distance' is how far patients say they can walk before the pain comes on. The 'total walking distance' is how far they can walk before the pain is so bad that they have to stop.

The pain is felt in major muscle groups and its location depends on the level at which the arteries are diseased. The calf muscle is most commonly affected and this is due to femoropopliteal disease, while pain in the thigh suggests common femoral or aortoiliac obstruction. Male patients who have bilateral common iliac or internal iliac artery occlusion may develop Leriche's syndrome, involving buttock claudication and erectile dysfunction.

Claudication is not in itself limb-threatening, although it is a marker for widespread atherosclerotic disease. With best medical therapy and supervised exercise programmes, 50% will improve, 30% will remain stable and only 20% will deteriorate further.

It is important to distinguish claudication due to arterial insufficiency from other causes of lower limb pain, which include

4.22 Fontaine classification of lower limb ischaemia

Stage	Description
I	Asymptomatic
II	Intermittent claudication
III	Night/rest pain
IV	Tissue loss (ulceration/gangrene)

4.23 The clinical features of arterial, neurogenic and venous claudication

	Arterial	Neurogenic	Venous
Pathology	Stenosis or occlusion of major lower limb arteries	Lumbar nerve root or cauda equina compression (spinal stenosis)	Obstruction to the venous outflow of the leg due to iliofemoral venous occlusion
Site of pain	Muscles, usually the calf but may involve thigh and buttocks	Ill-defined Whole leg May be associated with numbness and tingling	Whole leg 'Bursting' in nature
Laterality	Unilateral or bilateral	Often bilateral	Nearly always unilateral
Onset	Gradual after walking the 'claudication distance'	Often immediate on walking or standing up	Gradual, from the moment walking starts
Relieving features	On stopping walking, the pain disappears completely in 1–2 minutes	Bending forwards and stopping walking Patient may sit down for full relief	Leg elevation
Colour	Normal or pale	Normal	Cyanosed Often visible varicose veins
Temperature	Normal or cool	Normal	Normal or increased
Oedema	Absent	Absent	Always present
Pulses	Reduced or absent	Normal	Present but may be difficult to feel owing to oedema
Straight-leg raising	Normal	May be limited	Normal

osteoarthritis, neurogenic claudication and venous claudication (Box 4.23).

Any intervention for claudication is purely on the basis of symptomatic relief, since only a small minority of patients progress to critical limb ischaemia. The patient's age, occupation and comorbidities are important in determining the extent to which claudication limits their lifestyle. A postal worker who can walk only 200 metres has a serious problem but an elderly person who simply wants to cross the road to the shops may cope well. While absolute distances are important, it may be more helpful to ask specific questions about how symptoms affect the patient's lifestyle:

- Can you walk to the clinic from the bus stop or car park without stopping?
- Can you do your own shopping?
- What are you unable to do because of the pain?

Night pain

The patient goes to bed and falls asleep, but is then woken 1–2 hours later with severe pain in the foot, usually in the instep. The pain is due to poor perfusion, resulting from loss of the beneficial effects of gravity on lying down, and from the reductions in heart rate, BP and cardiac output that occur when sleeping. Patients may find relief by hanging the leg out of bed or by getting up and walking around. On return to bed, however, the pain recurs and patients often choose to sleep in a chair. This leads to dependent oedema, increased interstitial tissue pressure, a further reduction in tissue perfusion and ultimately a worsening of the pain.

Rest pain

Rest pain occurs when blood flow is insufficient to meet the metabolic demands of the tissues, even at rest. Critical ischaemia is defined as rest pain (persisting for more than 2 weeks and requiring opiate analgesia) or tissue loss associated with an ankle pressure of <50 mmHg or a toe pressure of <30 mmHg.

4.24 Signs of acute limb ischaemia

- Pallor
- Pulselessness
- Perishing cold
- Paraesthesia
- Pain (worse when muscle squeezed)
- Paralysis

Rest or night pain indicates severe, multilevel, lower limb PAD and is a 'red flag' symptom that mandates urgent referral to a vascular surgeon, as failure to revascularise the leg usually leads to the development of tissue loss (gangrene, ulceration) and amputation.

In patients with diabetes it may be difficult to differentiate between rest pain and diabetic neuropathy, as both may be worse at night. Neuropathic pain may not be confined to the foot, is associated with burning and tingling, is not relieved by dependency and is accompanied by dysaesthesia (pain or uncomfortable sensations, sometimes described as burning, tingling or numbness). Many patients with neuropathy cannot even bear the pressure of bedclothes on their feet.

Tissue loss (ulceration and/or gangrene)

In patients with severe lower limb PAD, perfusion is inadequate to support the tissues, and areas of tissue loss (gangrene) develop at the tips of the digits, gradually spreading proximally. Furthermore, even trivial injuries do not heal and cause ulceration. Tissue loss often progresses rapidly and, without revascularisation, leads to amputation and/or death.

Acute limb ischaemia

The classical features of acute limb ischaemia are the 'six Ps' (Box 4.24). Paralysis (inability to move the toes/fingers) and paraesthesia (numbness or tingling over the forefoot or dorsum of the hand) are the most important and indicate severe ischaemia affecting nerve and muscle function. Muscle tenderness is a grave sign

4.25 Acute limb ischaemia: embolus versus thrombosis in situ

	Embolus	Thrombosis
Onset and severity	Acute (seconds or minutes), ischaemia profound (no pre-existing collaterals)	Insidious (hours or days), ischaemia less severe (pre-existing collaterals)
Embolic source	Present	Absent
Previous claudication	Absent	Present
Pulses in contralateral leg	Present	Often absent, reflecting widespread peripheral arterial disease
Diagnosis	Clinical	Angiography
Treatment	Embolectomy and anticoagulation	Medical, bypass surgery, catheter-directed thrombolysis

indicating impending muscle infarction. A limb with these features will usually become irreversibly damaged unless the circulation is restored within a few hours.

It is important to consider the most likely underlying cause:

- Thromboembolism: usually from the left atrium in association with atrial fibrillation or myocardial infarction. There is usually no history of claudication.
- Thrombosis in situ: thrombotic occlusion of an already narrowed atherosclerotic arterial segment (Box 4.25). In this situation the patient is likely to have a past history of claudication.

Compartment syndrome

The perfusion pressure of a muscle is the difference between the mean arterial pressure and the pressure within the fascial compartment within which it lies. Compartment syndrome occurs where there is increased pressure within the fascial compartments of the limb that compromises the perfusion and viability of muscle and nerves. The calf is most commonly affected and the two leading causes are lower limb trauma (such as fractured tibia) and reperfusion injury following treatment of acute lower limb ischaemia. Failure to recognise and treat compartment syndrome may result in limb amputation. The key symptom is severe pain that is often unrelieved by opioids and exacerbated by active or passive movement. It is worth remembering that peripheral pulses are usually present, since the major arteries lie outside the fascial compartments and are not affected by increased compartment pressure. Compartment monitors can be used as an adjunct to measure compartment pressure.

Abdominal pain

Mesenteric ischaemia

Because of the rich collateral circulation of the gut, usually two of the three major visceral arteries (coeliac trunk, superior and inferior mesenteric arteries) must be critically stenosed or occluded before symptoms and signs of chronic mesenteric arterial insufficiency occur. Severe central abdominal pain typically develops 10–15 minutes after eating. The patient becomes scared of eating and significant weight loss is a universal finding. Diarrhoea may also be present and the non-specific nature of symptoms may result in misdiagnosis; the patient may have had numerous investigations, including laparotomy, before the diagnosis is made.

Acute mesenteric ischaemia is a surgical emergency. It is most commonly caused by an embolus from the heart or by thrombosis in situ of a pre-existing atherosclerotic plaque in one of the mesenteric vessels. It is often hard to diagnose in the early stages, as patients typically present initially with severe abdominal pain that is out of proportion to often unimpressive abdominal signs. Presentation with severe abdominal pain, shock, bloody diarrhoea and profound metabolic acidosis indicates infarction of the bowel, which carries a high mortality rate. Rarely, renal angle pain occurs from renal infarction or ischaemia, and is associated with visible or non-visible haematuria.

Any patient suspected of having visceral ischaemia should undergo urgent CT angiography.

Abdominal aortic aneurysm

Abdominal aortic aneurysm (AAA) is an abnormal focal dilatation of the aorta to at least 150% of its normal diameter (Fig. 4.32). It is often diagnosed incidentally during a CT scan or alternative imaging for other reasons.

Patients may present with abdominal and/or back pain, or occasionally with more subtle signs such as an awareness of abdominal pulsation or observation of ripples in the water when they are in the bath (wave sign). However, most patients are asymptomatic until the aneurysm ruptures.

The classical features of AAA rupture include abdominal/back pain, pulsatile abdominal mass, syncope and shock (hypotension), but these are not always present and it is important to have a low threshold of suspicion and consider early referral and/or CT imaging.

Digital ischaemia

Blue toes

Blue toe syndrome occurs when there is atheroembolism from an AAA or alternative proximal embolic source (such as popliteal aneurysm or atherosclerotic plaque). Patchy bluish discoloration appears over the toes and forefoot of one or both feet. There is usually a full set of pedal pulses. Although seemingly innocuous, this symptom should be taken seriously, as small emboli may herald the risk of a major embolus leading to acute limb ischaemia and even limb loss.

Vasospastic symptoms

Raynaud's phenomenon is digital ischaemia induced by cold and emotion. It has three phases (Fig. 4.33):

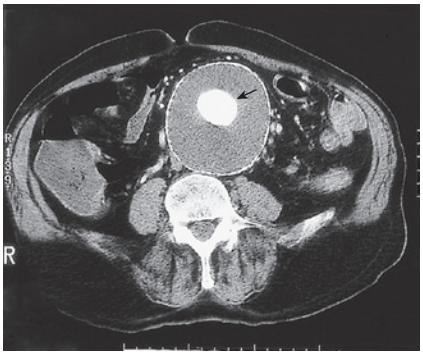
- pallor: due to digital artery spasm and/or obstruction
- cyanosis: due to deoxygenation of static venous blood (this phase may be absent)
- redness: due to reactive hyperaemia.

Raynaud's phenomenon may be primary (Raynaud's disease) and caused by idiopathic digital artery vasospasm, or secondary to other conditions (Raynaud's syndrome) such as drugs, connective tissue disease, hyperviscosity syndromes or use of power tools. While for most patients this is a self-limiting condition, a small minority develop tissue loss.

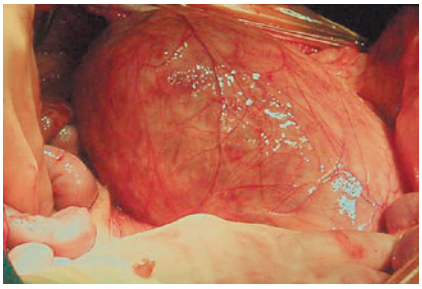
Patients over 40 years old presenting with unilateral Raynaud's phenomenon should be investigated for underlying PAD, especially if they have cardiovascular risk factors, diabetes or a smoking habit.



A



B



C

Fig. 4.32 Abdominal aortic aneurysm. A Abdominal X-ray showing calcification *(arrow)*. B Computed tomogram of the abdomen showing an abdominal aortic aneurysm *(arrow)*. C At laparotomy the aorta is seen to be grossly and irregularly dilated.

Stroke

Stroke is a focal neurological deficit that has a vascular cause and is discussed on page 123.

Past medical history

Is the patient known to have established peripheral vascular disease? Ask about previous investigations, operations or procedures. Is there a history of other atherosclerotic conditions such as coronary artery disease or cerebrovascular disease? Ask about risk factors for atherosclerotic disease, including hypertension, hypercholesterolaemia and diabetes mellitus. Are there any other comorbidities (such as severe cardiac or lung disease) that would make any potential operative intervention high-risk or futile?



A



B

Fig. 4.33 Raynaud's syndrome. A The acute phase, showing severe blanching of the tip of one finger. B Raynaud's syndrome occasionally progresses to fingertip ulceration or even gangrene. *(A and B) From Forbes CD, Jackson WF. Color Atlas of Clinical Medicine. 3rd edn. Edinburgh: Mosby; 2003.*

Drug history

Enquire about medication used for secondary prevention and adherence to these: antiplatelet, lipid-lowering, antihypertensive and diabetes therapies. Patients may be taking vasoactive drugs for claudication (naftidrofuryl or cilostazol, for example), although their efficacy in this setting is not clear. Enquire about other cardiac medications, as these may make symptoms of rest pain worse through their BP-lowering or negatively inotropic effects.

Family history

Ask about a family history of premature coronary or other vascular disease (p. 45). There is a strong familial association for AAAs so, where relevant, a family history should be sought.

Social history

Take a smoking history (p. 14). Enquire about occupation and activities of daily living. How are the patient's symptoms impacting on quality of life or employment?

The physical examination

Follow the routine described for the heart, looking for evidence of anaemia or cyanosis, signs of heart failure, and direct or

4.26 Signs suggesting vascular disease

Sign	Implication
Hands and arms	
Tobacco stains	Smoking
Purple discoloration of the fingertips	Atheroembolism from a proximal subclavian aneurysm
Pits and healed scars in the finger pulps	Secondary Raynaud's syndrome
Calcinosis and visible nail-fold capillary loops	Systemic sclerosis and CREST (calcinosis, Raynaud's phenomenon, oesophageal dysfunction, sclerodactyly, telangiectasia)
Wasting of the small muscles of the hand	Thoracic outlet syndrome
Face and neck	
Corneal arcus and xanthelasma	Hypercholesterolaemia
Horner's syndrome	Carotid artery dissection or aneurysm
Hoarseness of the voice and 'bovine' cough	Recurrent laryngeal nerve palsy from a thoracic aortic aneurysm
Prominent veins in the neck, shoulder and anterior chest	Axillary/subclavian vein occlusion
Abdomen	
Epigastric/umbilical pulsation	Aortoiliac aneurysm
Mottling of the abdomen	Ruptured abdominal aortic aneurysm or saddle embolism occluding aortic bifurcation
Evidence of weight loss	Visceral ischaemia

indirect evidence of PAD. Box 4.26 lists some of the direct and indirect signs of peripheral arterial disease.

Perform a detailed examination of the arterial pulses.

Examination sequence

Work down the body, starting with the hands, and using the sequence and principles of inspection, palpation and auscultation for each area.

Arms

- Examine the radial and brachial pulses (p. 47 and see Fig. 4.7).
- Measure the BP in both arms (p. 49 and see Fig. 4.12).

Neck

- Examine the carotid pulses (p. 47 and see Fig. 4.7).

Abdomen

- Inspect from the side for obvious pulsation.
- Palpate over the abdominal aorta. The aortic bifurcation is at the level of the umbilicus, so feel in the epigastrium for a palpable AAA. If you do feel an aneurysm, try to gauge its approximate size by placing the fingers of each hand on either side of it.
- In thin patients a tortuous but normal-diameter aorta can be palpable, but an aneurysm tends to be expansile rather than just pulsatile. A pulsatile mass below the umbilicus suggests an iliac aneurysm.

Clinical examination is unreliable in the detection of AAA since the aorta lies right at the back of the abdominal cavity. If you harbour any suspicion that the patient has an aneurysm, organise an ultrasound scan to confirm or exclude its presence.

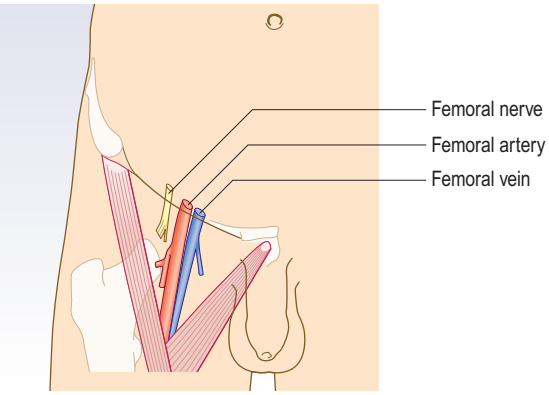


Fig. 4.34 Femoral triangle: vessels and nerves.

Legs

- Inspect and feel the legs and feet for changes of ischaemia, including temperature and colour changes (see Box 4.23), thin skin, brittle nails and absence of hair.
- Note scars from previous vascular or non-vascular surgery.
- Note the position, margin, depth and colour of any ulceration.
- Look specifically between the toes for ulcers and at the heels for ischaemic changes (the most common site of 'pressure sores').

Femoral pulse

- Ask the patient to lie down and explain what you are going to do.
- Place the pads of your index and middle fingers over the femoral artery. If you are having trouble feeling it (in an obese patient, for example), remember that the femoral artery lies at the mid-inguinal point, halfway between the anterior superior iliac spine and the pubic symphysis (Fig. 4.34).
- Remember that, while it is possible to listen for femoral bruits using the stethoscope diaphragm, the presence or absence of a bruit bears no relation to the likelihood or severity of aortoiliac disease.
- Palpate the femoral and radial pulses simultaneously to assess for radiofemoral delay (Fig. 4.35A).

Popliteal pulse

- With the patient lying down, flex their knee to 30 degrees.
- With your thumbs on the tibial tuberosity and your fingers in the midline posteriorly (2–3 cm below the skin crease), try to compress the artery against the back of the tibia (Fig. 4.35B).

The popliteal artery is usually hard to feel, so if it is readily palpable consider that there may be a popliteal artery aneurysm.

Posterior tibial pulse

- Feel 2 cm below and 2 cm behind the medial malleolus, using the pads of your middle three fingers (Fig. 4.35C).

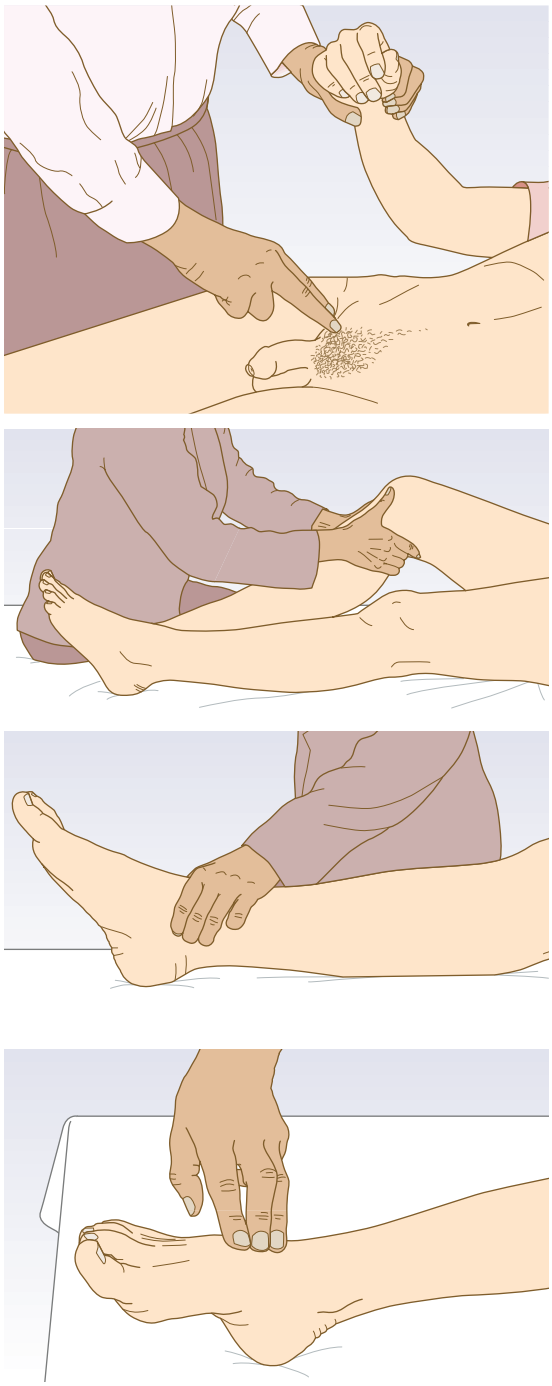


Fig. 4.35 Examination of the femoral, popliteal, posterior tibial and dorsalis pedis arteries. A Examine the femoral artery, while simultaneously checking for radiofemoral delay. B Feel the popliteal artery with your fingertips, having curled the fingers into the popliteal fossa. C Examine the posterior tibial artery. D Examine the dorsalis pedis artery.

Dorsalis pedis pulse

- Using the pads of your middle three fingers, feel at the origin of the first web space just lateral to the tendon of extensor hallucis longus (Fig. 4.35D).

The presence of foot pulses does not completely exclude significant lower limb PAD but they are almost always diminished or absent. If the history is convincing but pulses are felt, ask the patient to walk on a treadmill until pain develops; if they have flow-limiting PAD, their pulses will disappear and their APBI will fall (see below).

Buerger's test

Examination sequence

- With the patient lying supine, stand at the foot of the bed. Raise the patient's feet and support the legs at 45 degrees to the horizontal for 2–3 minutes.
- Watch for pallor with emptying and 'guttering' of the superficial veins.
- Ask the patient to sit up and hang their legs over the edge of the bed.
- Watch for reactive hyperaemia on dependency due to accumulation of vasoactive metabolites; the loss of pallor and spreading redness make a positive test.

Ankle : brachial pressure index

Assessing pulse status can be unreliable in patients with obesity or oedema. Routinely measure the ankle : brachial pressure index (ABPI) whenever there is difficulty palpating lower limb pulses or when PAD is suspected on the basis of the history.

Examination sequence

- Use a hand-held Doppler probe and a sphygmomanometer.
- Hold the probe over the posterior tibial artery at an angle of 45 degrees.
- Inflate a BP cuff round the ankle.
- Note the pressure at which the Doppler signal disappears. This is the systolic pressure in that artery as it passes under the cuff.
- Repeat, holding the probe over the dorsalis pedis artery.
- Measure the brachial BP in both arms, holding the Doppler probe over the brachial artery at the elbow or the radial artery at the wrist.

The ratio of the highest pedal artery pressure to the highest brachial artery pressure is the ABPI. In health, the ABPI is >1.0 when the patient is supine. An ABPI of <0.9 would be consistent with intermittent claudication and a value <0.4 may indicate critical limb ischaemia. Patients with critical limb ischaemia (rest pain, tissue loss) typically have an ankle BP of <50 mmHg and a positive Buerger's test.

Patients with lower limb PAD, particularly those with diabetes mellitus, often have incompressible, calcified crural arteries that give falsely reassuring pedal pressures and ABPI. Toe pressures may be more accurate in these patients and can be measured using a cuff round the base of the hallux with a laser Doppler probe at the tip of the toe.

Investigations

Further investigations must be carefully selected to provide the most information with the least risk to the patient and at least expense. Duplex ultrasound is often the first-line investigation of choice for unilateral disease, while bilateral symptoms can be investigated using a CT or MR angiogram (Box 4.27).

4.27 Investigations in peripheral arterial disease	
Investigation	**Indication/comment**
Duplex ultrasound	Carotid artery stenosis, abdominal aortic aneurysm surveillance, peripheral arterial disease
Computed tomography	Abdominal aortic aneurysm, peripheral arterial disease, carotid artery stenosis
Magnetic resonance imaging	Peripheral arterial disease, carotid artery stenosis, arteriovenous malformations
Angiography	Acute and chronic limb ischaemia, carotid artery stenosis Invasive angiography has largely been replaced by computed tomography/magnetic resonance angiography as a diagnostic test

PERIPHERAL VENOUS SYSTEM

Anatomy and physiology

Blood is returned to the heart from the peripheries by a network of deep (90%) and superficial (10%) veins. Venous return from the head and neck is passive, while blood from the legs must be pumped actively back up to the heart against gravity. Pressure on the sole of the foot on walking, together with contraction of muscles in the calf (the 'calf muscle pump') and, to a lesser extent, in the thighs and buttocks, drives blood back up through the veins. Backward flow (reflux) is prevented by valves that divide the long column of blood from the foot to the right atrium into a series of short, low-pressure segments. As a result, the 'ambulatory venous pressure' in the feet in health is usually <20 mmHg.

Deep veins follow the course of the main arteries. Valvular insufficiency causing venous reflux may be primary or post-thrombotic (following deep vein thrombosis, DVT). Following DVT, the vein may remain occluded or recanalise; valve function is usually compromised, however. The combination of deep venous obstruction and reflux leads to the signs and symptoms of post-thrombotic syndrome, including pain, venous claudication, blue discoloration, swelling, dilated superficial veins, skin changes and ulceration.

The long and short saphenous veins are the superficial veins of the lower limb and may also be affected by primary valvular failure and by valvular failure secondary to superficial thrombophlebitis. The long (great) saphenous vein passes anterior to the medial malleolus at the ankle, then up the medial aspect of the calf and thigh to join the common femoral vein in the groin at the saphenofemoral junction (Fig. 4.36).

The short (lesser) saphenous vein passes behind the lateral malleolus at the ankle and up the posterior aspect of the calf. It commonly joins the popliteal vein at the saphenopopliteal junction, which usually lies 2 cm above the posterior knee crease.

There are numerous intercommunications between the long and short saphenous veins, and between the deep and superficial venous systems, via perforator or communicating veins. The venous anatomy of the lower limb is highly variable.

The history

Common presenting symptoms

Lower limb venous disease presents in four ways:

- varicose veins
- deep venous thrombosis
- chronic venous insufficiency and ulceration
- superficial thrombophlebitis.

The severity of symptoms and signs may bear little relationship to the severity of the underlying pathology and the physical signs. Life-threatening DVT may be asymptomatic, while apparently trivial varicose veins may be associated with significant symptoms.

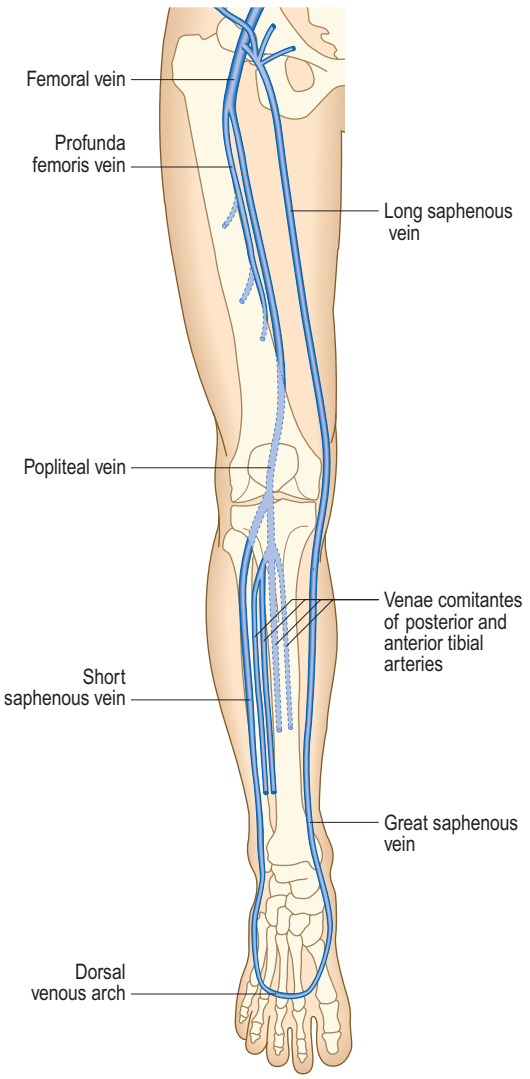


Fig. 4.36 Veins of the lower limb.

Pain

Patients with uncomplicated varicose (dilated, tortuous, superficial) veins often complain of aching leg discomfort, itching and a feeling of swelling (Fig. 4.37A). Symptoms are aggravated by prolonged standing and are often worse towards the end of the day. Once established, DVT causes pain and tenderness in the affected part (usually the calf). Superficial thrombophlebitis produces a red, painful area on the skin overlying the vein involved, and the vein may be palpable as a tender cord. Varicose ulceration may be surprisingly painless; if there is pain, this may be relieved by limb elevation but it is extremely important to exclude coexisting arterial disease (Box 4.28). Graduated compression bandaging is the mainstay of treatment for a venous leg ulcer,



A B

Fig. 4.37 Lower limb venous disease. A Varicose veins and associated haemosiderin deposition. B Venous ulcer. *(A) From Metcalfe M, Baker D. Varicose veins. Surgery (Oxford) 2008; 26(1):4–7.*

4.28 Clinical features of venous and arterial ulceration

Clinical feature	Venous ulceration	Arterial ulceration	Neuropathic ulceration
Sex	More common in women	More common in men	Equal in men and women
Risk factors	Thrombophilia, family history, previous deep vein thrombosis, varicose veins	Known peripheral vascular disease or risk factors for atherosclerotic disease, e.g. smoking, diabetes, dyslipidaemia, hypertension	Diabetes or other peripheral neuropathy (loss of sensation, loss of intrinsic foot muscle function, autonomic dysregulation)
Pain	Often painless but some patients have some pain that improves with elevating the leg	Severe pain, except in diabetics with neuropathy; improves on dependency	Painless or neuropathic pain
Site	Gaiter areas; 80% medial (long saphenous vein), 20% lateral (short saphenous vein)	Pressure areas (malleoli, heel, fifth metatarsal base, metatarsal heads and toes)	Pressure areas, sole of foot, tips of toes
Appearance	Shallow, irregular margin Slough on granulating base	Regular, 'punched out' Sloughy or necrotic base	Macerated, moist white skin surrounded by callus, often on load-bearing aspects (motor neuropathy)
Surrounding skin	Lipodermatosclerosis always present Oedema	Shiny, hairless, trophic changes	Dry due to reduced sweating (autonomic neuropathy)
Veins	Full and usually varicose	Empty with 'guttering' on elevation	Normal
Temperature	Warm Palpable pulses	Cold Absent pulses	Warm or cold due to autonomic neuropathy Palpable pulses

4.29 Risk factors for deep vein thrombosis

- Obesity
- Smoking
- Recent bed rest or operations (especially to the leg, pelvis or abdomen)
- Recent travel, especially long flights
- Previous trauma to the leg, especially long-bone fractures, plaster of Paris splintage and immobilisation
- Pregnancy or features suggesting pelvic disease
- Malignant disease
- Previous deep vein thrombosis
- Family history of thrombosis
- Inherited thrombophilia, e.g. factor V Leiden
- Recent central venous catheterisation, injection of drug
- Use of oral contraceptive or hormone replacement therapy

but is contraindicated unless there is documented evidence of adequate arterial circulation, which is assessed by feeling the pulses or by measuring the ABPI (see earlier).

Limb swelling

Swelling, or a feeling of swelling, even in the absence of visible signs, may be associated with lower limb venous disease. Enquire about risk factors for DVT (Box 4.29).

In upper limb DVT the arm is swollen and the skin is cyanosed and mottled, especially when dependent. Look for superficial distended veins (acting as collaterals) in the upper arm, over the shoulder region and on the anterior chest wall (Fig. 4.38). Symptoms are often exacerbated by activity, especially when holding the arm overhead.

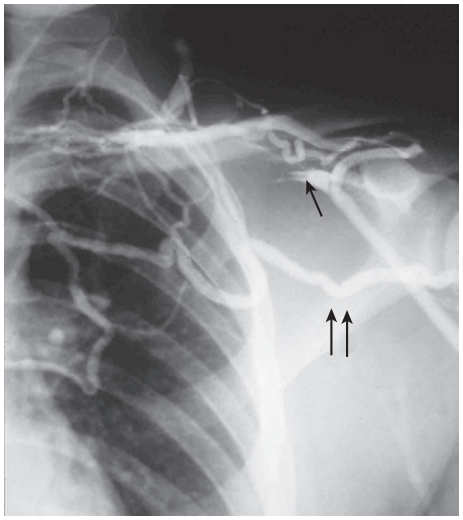
There may be a history of repetitive trauma at the thoracic outlet due to vigorous, repetitive exercise (swimming, weight lifting, racquet sports). Upper limb DVT may also complicate indwelling subclavian/jugular venous catheters.

Skin changes

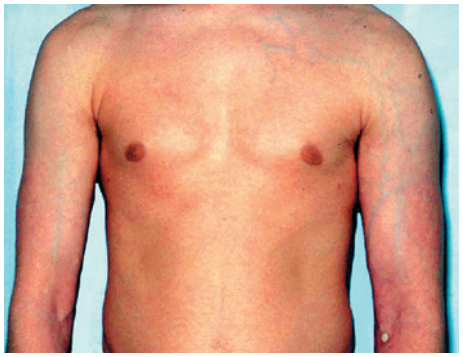
Chronic venous insufficiency is often associated with bluish discoloration of the distal extremity. A range of skin changes may be observed (see Fig. 4.37A). Varicose eczema leads to red, itchy, dry areas of skin over the lower leg. Venous hypertension causes extravasation of blood components into surrounding tissues, leading to haemosiderin deposition, which is seen as a brown discoloration of the skin, primarily around the medial aspect of the lower third of the leg. Lipodermatosclerosis occurs when there is an inflammatory response to the haemosiderin and causes red/purple discoloration and induration of the skin. The thickened, fibrotic skin forms a tight band around the lower leg, giving the appearance of an inverted champagne bottle. In *atrophie blanche*, there are multiple, small, white, scarred areas within the affected skin.

Chronic venous ulceration

In developed countries, about 70–80% of lower limb ulceration is primarily due to venous disease. Other causes include pyoderma gangrenosum, syphilis, tuberculosis, leprosy (Hansen's disease), sickle cell disease and tropical conditions. Chronic venous ulceration (see Fig. 4.37B) usually affects the medial aspect of the calf. Ulcers are shallow and pink (granulation tissue) or yellow/green (slough) in colour, with an irregular margin, and are usually associated with other skin changes of chronic venous insufficiency (varicose eczema, lipodermatosclerosis).



A



B

Fig. 4.38 Axillary vein thrombosis. A Angiogram. *Single arrow* shows site of thrombosis. *Double arrows* show dilated collateral vessels. B Clinical appearance with swollen left arm and dilated superficial veins.

Superficial venous thrombophlebitis

This condition affects up to 10% of patients with severe varicose veins and is more common during pregnancy. Recurrent superficial venous thrombophlebitis, especially that affecting different areas sequentially and non-varicose veins, may be associated with underlying malignancy. It may propagate into the deep system, leading to DVT and pulmonary embolism.

Past history

Enquire about previous varicose vein surgery and risk factors for DVT (Box 4.29).

The physical examination

Examination sequence

Expose the patient's legs and examine them with the patient standing and then lying supine.

- Inspect the skin for colour changes, swelling and superficial venous dilatation and tortuosity (varicose veins).
- Feel for any temperature difference.
- Press gently with your fingertip over the tibia above the ankle for a few seconds and then see if your finger has left a pit (pitting oedema). Remember to avoid areas that might be tender such as around ulcers.
- If the leg is grossly swollen, press at a higher level to establish how far the oedema extends.
- If you find oedema, check the JVP (p. 52). If the JVP is raised, this suggests cardiac disease or pulmonary hypertension as a cause, especially if both legs are oedematous.

Investigations

Tests such as the tourniquet and the Trendelenburg tests, to assess for saphenofemoral valve incompetence, are now obsolete and have been replaced by hand-held Doppler. With the patient standing, ask them to put their weight on the contralateral foot and position the hand-held Doppler probe over the long saphenous vein or saphenofemoral junction (2 cm below and medial to the mid-inguinal point). Squeeze the calf muscle and listen for blood flowing up through the long saphenous vein. If the valves are competent, you will hear a brief backflow of blood (<0.5 s) only when you release the calf muscle, which is physiological as the valves close. If the valves are incompetent, you will hear the prolonged sound of blood refluxing back down the vein.

OSCE example 1: Chest pain history

Ms McLeod, 62 years old, presents to you with intermittent chest pain.

Please take a history

- Introduce yourself and clean your hands.
- Invite the patient to describe the presenting symptoms, using open questioning.
- Take a detailed history of the presenting symptoms, including the onset, duration, site, quality and severity of the pain and any aggravating or relieving factors, in particular the relationship to exertion. Determine the functional consequences and any change in the pattern of symptoms.
- Ask about relevant past history and vascular risk factors, including hypertension, diabetes and hyperlipidaemia.
- Enquire about drug history or intolerances, including preventative therapies.
- Ask about premature coronary artery disease in first-degree relatives.
- Take a social history, including occupation, smoking and alcohol.
- Conduct a systematic inquiry. In particular, is there associated palpitation, breathlessness, orthopnoea and ankle swelling, or has there been any bleeding?
- Ask about any other patient concerns.
- Thank the patient and clean your hands.

Summarise your findings

Ms McLeod gives a 6-week history of intermittent chest discomfort. She reports a dull central ache that does not radiate to the arms or jaw. It occurs predominantly with effort, is worse on inclines or walking on cold mornings, and resolves at rest after a few minutes. These symptoms make her work as a carer challenging. She has no previous cardiac problems but is known to have hypertension and type 2 diabetes; she takes metformin for the latter. There is no family history of premature coronary artery disease and she has never been a smoker.

Suggest a likely diagnosis

The likely diagnosis is stable angina pectoris.

Suggest further evaluation

Full cardiovascular examination, blood glucose and lipid profile, and a 12-lead electrocardiogram. Consider referral for an exercise tolerance test or coronary angiogram.

OSCE example 2: Cardiac examination

Mr Munro, 82 years old, presents with progressive breathlessness and lightheadedness on exertion.

Please examine his cardiovascular system

- Introduce yourself and clean your hands.
- Carry out general observations. Is the patient tachypnoeic or distressed at rest? Are his hands cool?
- Measure the pulse, blood pressure and jugular venous pressure. Are the pulse volume and systolic pressure reduced? Is the jugular venous pressure elevated?
- Palpate the precordium. Is the apex more forceful or displaced?
- Auscultate over the apex and lower left, upper right and upper left sternal borders for the character of the first and second heart sounds, and the presence and characteristics of any added sounds or murmurs. If a murmur is heard, is there any radiation?
- Examine the chest, sacrum and lower limbs for signs of heart failure.
- Thank the patient and clean your hands.

Summarise your findings

Mr Munro appears comfortable. His heart rate is 80 bpm with a low-volume, slow-rising pulse. His blood pressure is 110/60 mmHg. The jugular venous pressure is not raised. There is an apical heave but the apex is not displaced. The first heart sound is normal but the second is diminished. There is a grade 3 ejection systolic murmur, loudest over the aortic area but heard widely, radiating to the carotids. There are fine end-inspiratory crackles at the lung bases.

Suggest a likely diagnosis

The likely diagnosis is aortic stenosis with left ventricular decompensation and heart failure.

Suggest initial investigations

Twelve-lead electrocardiogram, chest X-ray and transthoracic echocardiogram.

Integrated examination sequence for the cardiovascular system

- Position the patient: supine and reclined at 45 degrees, with the head resting on a pillow.
- Examine the general appearance:
  - Is the patient breathless, cyanosed, sweating or distressed?
  - Note body habitus (overweight or cachectic), Marfanoid features and the presence of radial or saphenous vein harvest scars.
- Check the hands, pulse and blood pressure, face and neck:
  - Hands: colour and temperature, tobacco staining, clubbing, splinter haemorrhages, Janeway lesions or Osler's nodes, tendon xanthomata.
  - Pulse: rate, rhythm, character and synchronicity of radial pulse, collapsing pulse, volume and character of brachial or carotid pulse.
  - Blood pressure: systolic and diastolic pressure at the brachial artery.
  - Face: central cyanosis, xanthelasmata, corneal arcus, petechiae.
  - Neck: timing, waveform and abnormalities of the jugular venous pressure, carotid bruits.
- Examine the precordium:
  - Inspection: look for midline sternotomy or left submammary scars, pacemaker site, visible pulsation.
  - Palpation: define the character and position of the apex beat, parasternal heave, thrills.
  - Auscultation: listen over the apex, lower left sternal border, upper right and left sternal borders, over the carotid arteries and left axilla. Listen with the patient on their left side and leaning forward during expiration.
    - Heart sounds: identify first and second heart sounds (S_1 and S_2), and any extra heart sounds (S_3 or S_4).
    - Additional sounds: clicks and snaps.
    - Murmurs in systole and/or diastole (timing, duration, character, pitch, intensity, location and radiation).
    - Pericardial rub.
- Other:
  - Listen for fine end-inspiratory crackles or pleural effusion at the lung bases.
  - Examine the abdomen for hepatomegaly or pulsatile liver.
  - Check for ankle and sacral oedema.

Peripheral arterial and venous system

- Inspection of the lower limbs:
  - Check temperature and colour, capillary refill time, skin discoloration, ulceration, varicosities, scars.
- Palpation:
  - Examine the abdomen for expansile aortic aneurysm.
  - Identify the femoral, popliteal, posterior tibial and dorsalis pedis pulses.
  - Identify pitting oedema.
  - Perform Buerger's test.
- Auscultation:
  - Listen for bruits over the abdomen and over the femoral arteries.